

Thesis Notes

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0.1 TODO

1. Read [Far06]
2. Read [CG13]
3. Read [DLS24]
4. ~~Ok so there's no continuous section in [DLS24], but what is the Schwarz genus? I.e. how many cases are there?~~
5. ~~Is it a good idea to just canonicalize via S_d^n ? Why or why not?~~
6. ~~What can we learn about $X = \{A \in \mathbb{R}^{d \times n} : \text{rank}(A) = d\}$ and $X/O(d)$ or $X/SO(d)$?~~
7. ~~Is a canonicalization up to homotopy sufficient?~~
8. ~~Use restrictions to bound sectional category on original space~~
9. ~~Consider $SO(d)$ acting on $\mathbb{R}^{d \times n}$~~
10. ~~Understand proof that there is no canonicalization of $(\mathbb{R}^{d \times n}, O(d))$ and $(\mathbb{R}^{d \times n}, SO(d))$.~~
11. ~~What if we want a *differentiable* section?~~
12. ~~Read up on frames~~
13. Read about equivariant neural networks/ML to provide advantages/disadvantages and motivation for canonicalization/frames
14. ~~Formalize Richter-Tchakaloff approach to creating canonicalization frames~~
 Generalize? I.e. we don't actually need to compute $\int_G f(g^{-1}x)dg$, just some invariant map.
15. Apply [DLS24]'s results on frames to [Ma+24]'s canonicalization frames
16. Consider probability measures on a (finite) group G as convex combinations of the simplex on G , and see what can be learned about weighted frames
17. What can we say about the cardinality of continuity-preserving frames for $O(d)$ and $SO(d)$ for $d > 2$?
18. Find (canonicalization) frames and/or bounds for (canonicalization) frame cardinality for more groups/spaces
19. Construct explicit piecewise continuous sections for:
 - $(\mathbb{R}^{d \times n}, O(d))$
 - $(\mathbb{R}^{d \times n}, SO(d))$
 - $(\mathbb{R}^{d \times n}, \Sigma_n)$
 - $(\mathbb{R}^{d \times n}/\Sigma_n, O(d))$
 - $(\mathbb{R}^{d \times n}/\Sigma_n, SO(d))$

20. Experiment computationally with piecewise continuous sections
21. Consider the case of rotations acting on images

Chapter 0

Introduction

0.1 Motivation

In machine learning, one is generally trying to learn a map $f : V \rightarrow W$ between vector spaces with some desired properties. For instance, if V is the vector space of tuples of age and height of some population and W represents the weights of that population, one may wish to identify a relationship between those three variables in the form of a map f taking tuples $(age, height) \in V$ to some $weight = f(age, height) \in W$. Generally, we want this map to reflect our intuition of how patterns in nature behave, meaning this map should at the very least be continuous (or nearly so), and potentially respect other properties or structure of the data.

A natural example of additional structure is when the data have some sort of inherent symmetry. This could arise when trying to label the pose of a person, for example, where any horizontal rotation of a person in a certain pose still is shaped in that pose. Mathematically, we can represent this with a group action on our space V by some group G of symmetries. In the case of poses, we might wish to construct a map f that ignores the actions our group G of horizontal rotations, meaning that every rotation is identified as the same pose. Formally, this is called an *invariant* or *G -invariant* map, defined to mean that $f(g \cdot v) = f(v)$ for all $g \in G$ and $v \in V$. Similarly, one may also wish to consider the case where W is also equipped with a G -action (for the same group G), in which case we may want a *(G -)equivariant* map $f : V \rightarrow W$, meaning $f(g \cdot v) = g \cdot f(v)$.

For practical purposes, one must then consider how to construct such a map. A very natural way is to consider a so-called *canonicalization*, where one first associates each instance of the data to a canonical representative respecting the symmetry [DLS24]. Concretely, if someone hands you a collection of points representing the joints of a human in a certain pose, you may attempt to rotate this collection of points so that the person would be facing you, then proceed to identify the pose. Mathematically, a canonicalization is then a map $c : V \rightarrow V$ such that $c(g \cdot v) = c(v) \in Gv$ for all $g \in G$ and $v \in V$. Then any continuous map $f : V \rightarrow W$, not necessarily invariant on its own, becomes an invariant map when composed with c , since $f(c(g \cdot v)) = f(c(v))$. The question then becomes whether or not such a canonicalization even exists. This question is addressed in [DLS24] and chapter 1.

Another rather intuitive way to construct such a map would be to take an average of some map over all possible symmetries. Informally, this would look like

$$\frac{1}{|G|} \sum_{g \in G} f(g \cdot v).$$

This of course doesn't make sense if the group G is infinite (like the case of rotational symmetries), but it can be made precise using integration and measure theory; this is explored further in [DLS24], [Pun+22], [Ma+24], and chapter 2.

1. GNNs can't fully distinguish graphs and can be computationally intensive (see references in introduction of [Pun+22])

0.2 Notation

- I use $A \cong B$ to denote that there exists an isomorphism between A and B in whatever the appropriate category is. For instance if X and Y are topological spaces then $X \cong Y$ means X and Y are homeomorphic, or if G and H are groups then $G \cong H$ means G and H are isomorphic as groups.
- For a homotopy $h : X \times [0, 1] \rightarrow Y$, I may also write $h_t : X \rightarrow Y$ with $h_t(x) := h(x, t)$.
- For any set X , I write c_x to mean the constant function at $x \in X$.
- Given a path $\gamma : [0, 1] \rightarrow X$, I write $\bar{\gamma}$ to mean the reversed path $\bar{\gamma}(t) := \gamma(1 - t)$.
- For any two paths $\alpha, \beta : [0, 1] \rightarrow X$ with $\alpha(1) = \beta(0)$, I write $\beta * \alpha$ to indicate their concatenation:

$$(\beta * \alpha)(t) = \begin{cases} \alpha(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ \beta(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}.$$

The exact parameterization is unimportant, but if it ever matters this is what I mean.

- Unless specified otherwise, for any group G I will write $e \in G$ to be the identity in G . If there is the potential for confusion between different groups, I will use subscripts to differentiate (e.g. e_G is the identity in G , e_H is the identity in H).
- I write $\mathbb{R}_{\text{rank } k}^{d \times n}$ to mean the set $\{A \in \mathbb{R}^{d \times n} : \text{rank}(A) = k\}$ and set $\mathbb{R}_*^{d \times n} = \mathbb{R}_{\text{rank } d}^{d \times n}$.
- Given some matrix $A \in \mathbb{R}^{n \times m}$, I write $A[r_1 : r_2, c_1 : c_2]$ to indicate the matrix of entries taken from rows r_1 through r_2 (inclusive) and columns c_1 through c_2 (inclusive) of A . If any value is left empty, I mean to include all the way to the first/last row/column depending on which entry is empty (e.g. $A[:, : 2]$ is the first two columns of A). For any programmers, this is just the Python library `numpy`'s syntax.

Chapter 1

Canonicalization

1.1 Background

1.1.1 Background from [Far06]

Definition 1.1.1. For fixed $a = (a_1, \dots, a_m) \in \mathbb{R}_+^m$, we define the variety

$$M(a) = \left\{ (z_1, \dots, z_m) \in (S^2)^m : \sum_{i=1}^m a_i z_i = 0 \right\} / SO(3)$$

where $SO(3)$ acts coordinate-wise on $(S^2)^m$. This variety describes all polygons in $\mathbb{R}^3$ with sidelengths $a_1, \dots, a_m$.

Definition 1.1.2. Let X be a path-connected topological space. Then $PX = C([0, 1]; X)$ is the space of all paths $\gamma : [0, 1] \rightarrow X$ equipped with the compact-open topology.

Recall that the compact-open topology has a subbase given by sets of the form

$$V(K, U) := \{f \in C(X, Y) : f[K] \subseteq U\}$$

for $K \subseteq [0, 1]$ compact and $U \subseteq X$ open [Wik26a].

Definition 1.1.3. For spaces E and B , a map $p : E \rightarrow B$ has the homotopy lifting property (HLP) for a space X if for every homotopy $h_t : X \rightarrow B$ and every lift $\tilde{h}_0 : X \rightarrow E$ of h_0 ($h_0 = p \circ \tilde{h}_0$), there exists a homotopy $\tilde{h}_t : X \rightarrow E$ that lifts h_t ($h_t = p \circ \tilde{h}_t$).

Definition 1.1.4. For spaces E and B , a map $p : E \rightarrow B$ is a (Hurewicz) fibration if p satisfies the homotopy lifting property for all topological spaces [Wik26c].

Remark 1.1.5. The map $\pi : PX \rightarrow X \times X$ given by $\pi(\gamma) = (\gamma(0), \gamma(1))$ is a fibration.

Proof. Let Y be any topological space and let $h_t : Y \rightarrow X \times X$ be a homotopy with some $\tilde{h}_0 : Y \rightarrow PX$ such that $h_0 = \pi \circ \tilde{h}_0$. Then we simply define

$$\tilde{h}_t(y)(s) = \begin{cases} (h_{t(1-3s)}(y))_1 & \text{if } 0 \leq s \leq \frac{1}{3} \\ \tilde{h}_0(y)(3(s - \frac{1}{3})) & \text{if } \frac{1}{3} \leq s \leq \frac{2}{3} \\ (h_{t(3s-2)}(y))_2 & \text{if } \frac{2}{3} \leq s \leq 1 \end{cases}.$$

Intuitively, this simply defines a path from $(h_t(y))_1$ to $(h_t(y))_2$ by following:

1. $(h_\bullet(y))_1$ from $(h_t(y))_1$ to $(h_0(y))_1$,
2. $\tilde{h}_0$ from $(h_0(y))_1 = \tilde{h}_0(y)(0)$ ($h_0 = \pi \circ \tilde{h}_0$) to $\tilde{h}_0(y)(1)$,
3. and $(h_\bullet(y))_2$ from $\tilde{h}_0(y)(1) = (h_0(y))_2$ ($\pi \circ \tilde{h}_0 = h_0$) to $(h_t(y))_2$.

By construction $\tilde{h}_t$ is continuous, so we conclude that π is a fibration. □

Definition 1.1.6. A motion planning algorithm is a section $s : X \times X \rightarrow PX$ of π , i.e. $\pi \circ s = \text{id}_{X \times X}$.

Lemma 1.1.7. A continuous motion planning algorithm in X exists if and only if X is contractible.

Proof. First assume that there exists a continuous motion planning algorithm, i.e. a continuous section $s : X \times X \rightarrow PX$ of π , and fix some $x_0 \in X$. Then for every $x \in X$, define $\gamma_x = s(x_0, x) \in PX$. Now define a homotopy $h_t : X \rightarrow X$ via $h_t(x) = \gamma_x(t)$. Observe that $h_0(x) = \gamma_x(0) = x_0$ and $h_1(x) = \gamma_x(1) = x$ so $h_0 = c_{x_0}$ and $h_1 = \text{id}_X$. Since s is continuous, this means that h is a homotopy from c_{x_0} to id_X and therefore X is contractible.

Now instead assume that X is contractible and let $h_t : X \rightarrow X$ be a homotopy exhibiting this ($h_0 = c_{x_0}$ for some $x_0 \in X$ and $h_1 = \text{id}_X$). Then for each $x \in X$, define $\gamma_x \in PX$ via $\gamma_x(t) = h_t(x)$ and define $s : X \times X \rightarrow PX$ by $s(x, y) = \gamma_y * \overline{\gamma_x}$. Then s is a section since $\pi(s(x, y)) = ((\gamma_y * \overline{\gamma_x})(0), (\gamma_y * \overline{\gamma_x})(1)) = (x, y)$, and s is continuous by construction. $\square$

Definition 1.1.8. A topological space X is a Euclidean Neighborhood Retract (ENR) if there exists an embedding $i : X \rightarrow \mathbb{R}^n$ for some n with an open neighborhood U of $i[X]$ such that U retracts onto $i[X]$.

Definition 1.1.9. A motion planning algorithm $s : X \times X \rightarrow PX$ is tame if there are finitely many disjoint sets $F_i \subseteq X \times X$ such that $X \times X = \cup_i F_i$, $s|_{F_i}$ is continuous, and F_i is an ENR.

Definition 1.1.10. The topological complexity of a tame motion planning algorithm is the minimum number of sets F_i one can take in definition 1.1.9.

Definition 1.1.11. For a path-connected space X , the topological complexity $\mathbf{TC}_1(X)$ is the minimum topological complexity of tame motion planning algorithms in X . We set $\mathbf{TC}_1(X) = \infty$ if X admits no tame motion planning algorithms.

Remark 1.1.12. $\mathbf{TC}_1(X) = 1$ if and only if X is a contractible ENR.

Proof. This trivially follows from lemma 1.1.7 and definitions 1.1.8 and 1.1.9. $\square$

Definition 1.1.13. The Schwarz genus or sectional category of a fibration $p : E \rightarrow B$, denoted $\text{secat}(p)$ is the minimum number k of open sets $U_1, \dots, U_k$ covering B such that each U_j admits a continuous section $s_j : U_j \rightarrow E$ of p .

Remark 1.1.14. The Schwarz genus of a fibration $p : E \rightarrow B$ is 1 if and only if p admits a continuous section $s : B \rightarrow E$.

Definition 1.1.15. For a path-connected pointed space (X, x_0) , we define

$$P_0X := \{\gamma \in C([0, 1]; X) : \gamma(0) = x_0\}.$$

Definition 1.1.16. For a path-connected pointed space (X, x_0) , the Lusternik-Schnirelmann category of X , denoted $\text{cat}(X)$, is the Schwarz genus of the fibration $\pi_0 : P_0X \rightarrow X, \gamma \mapsto \gamma(1)$.

Definition 1.1.17. For a path-connected space X , the topological complexity $\mathbf{TC}_2(X)$ is the Schwarz genus of the fibration $\pi : PX \rightarrow X \times X$.

Remark 1.1.18. $\mathbf{TC}_2(X) = 1$ if and only if X is contractible.

Proof. This is a direct corollary of lemma 1.1.7. $\square$

Lemma 1.1.19. *Let $p : E \rightarrow B$ be a fibration, and let $B' \subseteq B$ with $E' := p^{-1}[B']$ and $p' = p|_{E'} : E' \rightarrow B'$. Then $\text{secat}(p') \leq \text{secat}(p)$.*

Proof. Assume the $\text{secat}(p) < \infty$ and consider a corresponding minimal set of U_i 's with local sections $s_i : U_i \rightarrow E$ of p . Observe that $s_i|_{B'} : U_i \cap B' \rightarrow E'$ is then a local section of p' , so $\text{secat}(p') \leq \text{secat}(p)$. If $\text{secat}(p) = \infty$, we trivially conclude $\text{secat}(p') \leq \infty = \text{secat}(p)$, so in either case the Schwarz genus of p' is bounded above by that of p .¹ $\square$

Lemma 1.1.20. *For any fibration $p : E \rightarrow B$, $\text{secat}(p) \leq \text{cat}(B)$.*

*Proof.*² Take $\text{cat}(B)$ -many open sets U_i with continuous sections $s_i : U_i \rightarrow P_0B$ of $\pi_0 : P_0B \rightarrow B$. Observe that each s_i induces a homotopy $h_i : U_i \times [0, 1] \rightarrow B$ via $h_i(u, t) := s_i(u)(t)$. Then with the base point $b_0 \in B$, we can fix a corresponding element $e_0 \in p^{-1}[\{b_0\}] \subseteq E$ and define $\tilde{h}_i : U_i \times \{0\} \rightarrow E$ by $\tilde{h}_i(u, 0) = e_0$. This gives us that $p(\tilde{h}_i(u, 0)) = p(e_0) = b_0 = s_i(u)(0) = h_i(u, 0)$, so by the homotopy lifting property of p there exists a homotopy $\tilde{h}_i : U_i \times [0, 1] \rightarrow E$ such that $p \circ \tilde{h}_i = h_i$. Then observe we can define maps $r_i : U_i \rightarrow E$ via $r_i(u) := \tilde{h}_i(u, 1)$, giving us $p(r_i(u)) = p(\tilde{h}_i(u, 1)) = h_i(u, 1) = u$. We therefore conclude that each r_i is a continuous section of p and so $\text{secat}(p) \leq \text{cat}(B)$. $\square$

Proposition 1.1.21. *For a path-connected space X , $\text{cat}(X) \leq \mathbf{TC}_2(X) \leq \text{cat}(X \times X)$.*

Proof. We have the fibration $\pi : PX \rightarrow X \times X$ from before, as well as $\pi_0 : P_0X \rightarrow X \times X$ given by $\pi_0(\gamma) = (x_0, \gamma(1))$. Observe that $\text{cat}(X)$ is exactly the Schwarz genus of π_0 and $P_0X = \pi^{-1}[\{x_0\} \times X]$, so we immediately know by lemma ?? that $\text{cat}(X) \leq \mathbf{TC}_2(X)$. Then by lemma 1.1.20 we conclude $\mathbf{TC}_2(X) = \text{secat}(\pi : PX \rightarrow X \times X) \leq \text{cat}(X \times X)$. $\square$

Theorem 1.1.22. *$\mathbf{TC}_2$ is a homotopy invariant.*

Definition 1.1.23. *The order of instability of the composition in definition 1.1.9 is the maximum r such that there is some sequence of indices $1 \leq i_1 < \dots < i_r \leq k$ where $\overline{F_{i_1}} \cap \dots \cap \overline{F_{i_r}}$ is non-empty. The order of instability of a motion planning algorithm s is then the minimum order of instability of decompositions for s .*

Definition 1.1.24. *The topological complexity $\mathbf{TC}_3(X)$ is the minimum order of instability of all tame motion planning algorithms in X .*

Remark 1.1.25. $\mathbf{TC}_3(X) \leq \mathbf{TC}_1(X)$.

Proof. This immediately follows from definitions 1.1.11 and 1.1.24. $\square$

Definition 1.1.26. *A random n -valued path σ in X is a pair of n -tuples $(\gamma_1, \dots, \gamma_n) \in (PX)^n$ and $(p_1, \dots, p_n) \in [0, 1]^n$ such that all the γ_i 's have the same start and end points and $p_1 + \dots + p_n = 1$. We write σ as a formal linear combination $\sigma = p_1\gamma_1 + \dots + p_n\gamma_n$.*

We then define P_nX to be the set of all n -valued random paths in X with the subspace topology induced by $\bigoplus_n PX$, and we have the same fibration $\pi : P_nX \rightarrow X \times X$ as before taking n -valued random paths to their end points.

¹If one wishes to more precisely compare cardinalities, the reasoning in the finite case works.

²Consulted Claude for the basic idea in this proof.

Definition 1.1.27. An n -valued random motion planning algorithm is a continuous section $s : X \times X \rightarrow P_n X$ of π .

Observe that $s(x, y)$ is simply a probability distribution on n paths from x to y .

Definition 1.1.28. The topological complexity $\mathbf{TC}_4(X)$ is the minimum integer n such that there is some n -valued random motion planning algorithm $s : X \times X \rightarrow P_n X$.

Theorem 1.1.29. If X is a simplicial polyhedron, then

$$\mathbf{TC}_1(X) = \mathbf{TC}_2(X) = \mathbf{TC}_3(X) = \mathbf{TC}_4(X).$$

Proof sketch. The proof proceeds in steps:

- Step 1. $\mathbf{TC}_1(X) \leq \mathbf{TC}_2(X)$: Expand each F_i to a U_i that deformation retracts back onto it and use the deformation retract to extend $s|_{F_i} : F_i \rightarrow PX$ so $s_i : U_i \rightarrow PX$.
- Step 2. $\mathbf{TC}_2(X) \leq \mathbf{TC}_1(X)$: Trim each U_i just enough to get disjoint F_i 's that are ENRs.
- Step 3. $\mathbf{TC}_3(X) \leq \mathbf{TC}_2(X)$: Same basic idea as step 1.
- Step 4. $\mathbf{TC}_1(X) \leq \mathbf{TC}_3(X)$: Done already in remark 1.1.25.
- Step 5. $\mathbf{TC}_2(X) \leq \mathbf{TC}_4(X)$: Construct open sets $U_j \subseteq X \times X$ from the functions $p_j : X \times X \rightarrow [0, 1]$, and conclude by construction.
- Step 6. $\mathbf{TC}_4(X) \leq \mathbf{TC}_2(X)$: Extend the s_i 's arbitrarily, weight them via a partition of unity, and conclude.

□

Definition 1.1.30. Moving forward, we define the topological complexity of a space X to be $\mathbf{TC}(X) := \mathbf{TC}_2(X)$.

Definition 1.1.31. The order of an open cover $\mathcal{U}$ of X is the least integer n such that each point of X belongs to at most n sets of $\mathcal{U}$ (if such an integer exists) [Wik26d].

Definition 1.1.32. The covering dimension of a space X , denoted $\dim X$, to be the least integer n such that every finite open cover $\mathcal{U}$ of X has a refinement $\mathcal{V}$ of order $n + 1$. If no such n exists, then we set $\dim X = \infty$ [Wik26d].

Lemma 1.1.33. Let X be a path-connected paracompact locally contractible space. Then

$$\text{cat}(X) \leq \dim(X) + 1.$$

Lemma 1.1.34. Let X be a path-connected paracompact locally contractible space. Then

$$\dim(X \times X) = 2 \dim X.$$

Theorem 1.1.35. Let X be a path-connected paracompact locally contractible space. Then

$$\mathbf{TC}(X) \leq 2 \dim X + 1.$$

Proof. By lemmas 1.1.20, 1.1.33, and 1.1.34 we have

$$\mathbf{TC}(X) \leq \text{cat}(X \times X) \leq \dim(X \times X) + 1 = 2 \dim X + 1.$$

□

Theorem 1.1.36. *Let X be an r -connected CW-complex. Then*

$$\mathbf{TC}(X) \leq \frac{2 \dim X + 1}{r + 1} + 1.$$

Definition 1.1.37. *The tensor product $H^\bullet(X) \otimes H^\bullet(X)$ is a graded algebra with multiplication given by*

$$(u_1 \otimes v_1) \cdot (u_2 \otimes v_2) = (-1)^{|v_1||u_2|} (u_1 u_2) \otimes (v_1 v_2)$$

where $|x|$ denotes the degree of the cohomology class of x .

Definition 1.1.38. *The cup product $\smile: H^k(X) \otimes H^\ell(X) \rightarrow H^{k+\ell}(X)$ is given by*

$$(\alpha \smile \beta)(\sigma) := \alpha(\sigma_{[0, \dots, k]}) \cdot \beta(\sigma_{[k, \dots, k+\ell]}).$$

Note that this means $\smile$ induces a graded algebra $H^\bullet(X)$.

Definition 1.1.39. *The ideal of the zero-divisors of $H^\bullet(X)$ is the kernel of*

$$\smile: H^\bullet(X) \otimes H^\bullet(X) \rightarrow H^\bullet(X).$$

Definition 1.1.40. *The zero-divisors-cup-length or simply cup-length of $H^\bullet(X)$, denoted $\text{cup}(X)$ is the length of the longest nontrivial product in the ideal of zero-divisors of $H^\bullet(X)$.*

Theorem 1.1.41. *$\mathbf{TC}(X)$ is greater than the zero-divisors-cup-length of $H^\bullet(X)$.*

Example 1.1.42. *Consider S^n , letting $u \in H^n(S^n)$ be the fundamental class and $1 \in H^0(S^n)$ be the unit. Recall that $H^k(S^n) = 0$ for $k \neq 0, n$, and note $H^n(S^n) = \langle u \rangle$ and $H^0(S^n) = \langle 1 \rangle$. We then set $a = u \otimes 1 - 1 \otimes u$ and $b = u \otimes u$, and note that, up to coefficients, a and b are the only (potentially) interesting elements of $H^\bullet(S^n) \otimes H^\bullet(S^n)$. Observe that $u^2 \in H^{2n}(S^n) = 0$ so $u^2 = 0$ and we compute*

$$\begin{aligned} a \cdot a &= (u \otimes 1 - 1 \otimes u) \cdot (u \otimes 1 - 1 \otimes u) \\ &= (u \otimes 1) \cdot (u \otimes 1) - (u \otimes 1) \cdot (1 \otimes u) - (1 \otimes u) \cdot (u \otimes 1) + (1 \otimes u) \cdot (1 \otimes u) \\ &= (-1)^{|1||u|} (u^2 \otimes 1^2) - (-1)^{|1||1|} (u \otimes u) - (-1)^{|u||u|} (u \otimes u) + (-1)^{|u||1|} (1^2 \otimes u^2) \\ &= -u \otimes u - (-1)^{n^2} (u \otimes u) \\ &= ((-1)^{n+1} - 1) u \otimes u. \end{aligned}$$

So if n is even then $a^2 = -2b$ and if n is odd $a^2 = 0$, giving us that the zero-divisors-cup-length of $H^\bullet(S^n)$ is 1 for odd n and 2 for even n .

Theorem 1.1.43. *For spaces X and Y ,*

$$\mathbf{TC}(X \times Y) \leq \mathbf{TC}(X) + \mathbf{TC}(Y) - 1.$$

Corollary 1.1.44. *For spaces $X_1, \dots, X_n$ with $\mathbf{TC}(X_i) \leq M$ for all $1 \leq i \leq n$,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \leq nM - 1.$$

Theorem 1.1.45. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \geq n + 1.$$

Corollary 1.1.46. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n)$$

grows linearly with n .

Proof. This follows directly from corollary 1.1.44 and theorem 1.1.45. $\square$

Proposition 1.1.47. *For $U \subseteq X \times X$, there exists a section $s : U \rightarrow PX$ of $\pi : PX \rightarrow X \times X$ if and only if the inclusion $i : U \hookrightarrow X \times X$ is homotopic to some diagonal map $f : U \rightarrow X \times X$ (i.e. $f[U]$ is contained in the diagonal of $X \times X$).*

Proof. First assume we have a section $s : U \rightarrow PX$ of π and consider the diagonal map $f : U \rightarrow X \times X$ given by $f(x, y) := (y, y)$. Then we can construct a homotopy $h_t : U \rightarrow X \times X$ by $h_t(x, y) := (s(x, y)(t), y)$. Observe that $h_0(x, y) = (x, y) = i(x, y)$ and $h_1(x, y) = (y, y) = f(x, y)$ so we have a homotopy from $i : U \rightarrow X \times X$ to a diagonal map $f : U \rightarrow X \times X$.

Conversely, assume we have a homotopy $h_t : U \rightarrow X \times X$ such that $h_0 = i$ and $h_1 =: f$ is a diagonal map; denote $\hat{f} = f_1 = f_2$. Then for some $(x, y) \in U$, we have two paths, $\gamma_1(t) = (h_t(x, y))_1$ from x to $\hat{f}(x, y)$ and $\gamma_2(t) = (h_t(x, y))_2$ from y to $\hat{f}(x, y)$. Then we can define $s : U \rightarrow PX$ by $s(x, y) = \overline{\gamma_2} * \gamma_1$, and by construction we have $\pi \circ s = \text{id}$ $\square$

Corollary 1.1.48. *The topological complexity $\mathbf{TC}(X) = \mathbf{TC}_2(X)$ is the smallest integer k such that $X \times X$ can be covered by k open sets where each inclusion $U_j \hookrightarrow X \times X$ is homotopic to a diagonal map $f : U \rightarrow X \times X$.*

Proof. This follows immediately from definition 1.1.17 and proposition 1.1.47. $\square$

Proposition 1.1.49. *If X is an ENR then*

$$\mathbf{TC}(X) \geq \text{cat}((X \times X)/\Delta X) - 1$$

where $\Delta X \subseteq X \times X$ is the diagonal.

1.1.2 Background from [CG13]

Definition 1.1.50. The sectional category of a map $p : E \rightarrow B$, denoted $\text{secat}(p)$, is the least integer k such that B may be covered by k open sets U_j that each admit a map $s_j : U_j \rightarrow E$ where $p \circ s_j$ is homotopic to the inclusion $i_j : U_j \hookrightarrow B$.

Remark 1.1.51. For a fibration $p : E \rightarrow B$, definition 1.1.50 corresponds to definition 1.1.13 given in [Far06].

Proof. First, let U_j and $s_j : U_j \rightarrow E$ be as in definition 1.1.13, i.e. $p \circ s_j = i_j$. Then trivially $p \circ s_i \simeq i_j$.

Conversely, let U_j and $s_j : U_j \rightarrow E$ be as in definition 1.1.50, i.e. $p \circ s_j \simeq i_j$, and fix some $j \leq k$. Let $h_t : U_j \rightarrow B$ be the homotopy witnessing $p \circ s_j \simeq i_j$ with $h_0 = p \circ s_j$ and $h_1 = i_j$. We define the map $\tilde{h}_0 : U_j \rightarrow E$ by $\tilde{h}_0 = s_j$, and observe that by construction $p \circ \tilde{h}_0 = p \circ s_j = h_0$. Then by the homotopy lifting property we have a homotopy $\tilde{h}_t : U_j \rightarrow E$ such that $p \circ \tilde{h}_t = h_t$. In particular, this gives us a map $\hat{s}_j := \tilde{h}_1 : U_j \rightarrow E$ such that $p \circ \hat{s}_j = h_1 = i_j$. $\square$

Definition 1.1.52. The Lusternik-Schnirelmann category of a space X , denoted $\text{cat}(X)$, is the least integer k such that X may be covered by k open sets U_j where each inclusion $i_j : U_j \hookrightarrow X$ is null-homotopic.

Remark 1.1.53. For a path-connected space X , definition 1.1.52 corresponds to definition 1.1.16 given in [Far06]. Note that definition 1.1.16 only makes sense if X is path-connected anyway.

Proof. First, let $U_j \subseteq X$ and $s_j : U_j \rightarrow P_0X$ be as in definition 1.1.16, i.e. $\pi_0 \circ s_j = i_j$. We then simply define a homotopy $h_t : U_j \rightarrow X$ via $h_t(x) = s_j(x)(t)$, so $h_0(x) = s_j(x)(0) = x_0$ and $h_1(x) = s_j(x)(1) = x$. Therefore $h_0 = c_{x_0}$ and $h_1 = i_j$, so $c_{x_0} \simeq i_j$.

Conversely, let $U_j \subseteq X$ be as in definition 1.1.52 with $i_j \simeq c_{x_j}$ for some $x_j \in X$. Since X is path-connected all constant maps are homotopic, so without loss of generality each $x_j = x_0$. Fixing some j , let $h_t : U_j \rightarrow X$ be the homotopy witnessing $i_j \simeq c_{x_0}$ where $h_0 = c_{x_0}$ and $h_1 = i_j$. Then we define $s_j : U_j \rightarrow P_0X$ by $s_j(x)(t) = h_t(x)$, so

$$\pi_0(s_j(x)) = s_j(x)(1) = h_1(x) = i_j(x) = x.$$

$\square$

Proposition 1.1.54. For any spaces E and B along with a surjective map $p : E \rightarrow B$, $\text{secat}(p) \leq \text{cat}(B)$.

Proof. Take $\text{cat}(B)$ -many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \rightarrow B$. For each j , fix $e_j \in p^{-1}[c_j[U_j]]$ and define $s_j : U_j \rightarrow E$ by $s_j = c_{e_j}$. Observe that $p \circ s_j = c_j$ by definition, so $p \circ s_j \simeq i_j$. Therefore $\text{secat}(p) \leq \text{cat}(B)$. $\square$

Proposition 1.1.55. For a contractible space E and any space B along with a map $p : E \rightarrow B$, $\text{cat}(B) \leq \text{secat}(p)$.

Proof. Take $\text{secat}(p)$ -many open sets U_j with $s_j : U_j \rightarrow E$ such that $B = \cup_j U_j$ and $p \circ s_j \simeq i_j$. Since E is contractible we have that each map s_j is homotopic to a constant map c_j , and immediately this means that $p \circ s_j$ is homotopic to $p \circ c_j$ which is also constant. Therefore $i_j \simeq p \circ c_j$ and U_j is null-homotopic so $\text{cat}(B) \leq \text{secat}(p)$. $\square$

Corollary 1.1.56. *If E is a contractible space, B is any space, and $p : E \rightarrow B$ is a surjection, then $\text{secat}(p) = \text{cat}(B)$.*

Proof. This directly follows from propositions 1.1.54 and 1.1.55. $\square$

Definition 1.1.57. *Given a commutative ring R and an ideal $I \subseteq R$, the nilpotency of I , denoted $\text{nil } I$, is the maximum integer k such that we can find elements $r_1, \dots, r_k \in I$ with $r_1 \cdots r_k \neq 0$.*

Proposition 1.1.58. *Let $p : E \rightarrow B$ be a fibration and let $p^* : H^*(B) \rightarrow H^*(E)$ be the induced homomorphism on cohomology. Then $\text{secat}(p) > \text{nil ker } p^*$.*

Definition 1.1.59. *Let G be a topological group and X be a space, and assume we have some continuous G -action on X (i.e. $(g, x) \mapsto g \cdot x$ is continuous). Then X is a G -space.*

Definition 1.1.60. *For $x \in X$ a G -space, the isotropy group or stabilizer group of x is*

$$G_x := \{g \in G : gx = x\}.$$

Remark 1.1.61. G_x is closed for all $x \in X$.

Proof. Fix some $x \in X$ and consider a point $g \in \overline{G_x}$. Then there is a net $\{g_d\}_{d \in D} \subseteq G_x$ such that $g_d \rightarrow g$. Since each $g_d x = x$ by definition, by continuity we have that $g_d x \rightarrow gx = x$, so $g \in G_x$ and G_x is closed. $\square$

Definition 1.1.62. *Given a G -space X and $x \in X$, the orbit of x is*

$$Gx := \{gx : g \in G\}.$$

Remark 1.1.63. *The map $\phi : G/G_x \rightarrow Gx$ given by $\phi(gG_x) = gx$ is a homeomorphism.*

Proof. First, let $g, g' \in G$ such that $gG_x = g'G_x$; we wish to show that $gx = g'x$. Since $gG_x = g'G_x$, there exists some $h \in G_x$ such that $ge = g'h$. Then we have

$$gx = gex = g'hx = g'x.$$

So ϕ is well-defined.

To see that ϕ is continuous, consider the quotient map $q : G \rightarrow G/G_x$; we wish to show that $\phi \circ q : G \rightarrow Gx$ is continuous. Observe that $\phi(q(g)) = \phi(gG_x) = gx$, so it immediately follows that $\phi \circ q$ is continuous.

Lastly, to see that ϕ is an open map, consider some open set $UG_x \subseteq G/G_x$. Then $\phi[UG_x] = Ux$ which is clearly open.

Therefore we conclude that ϕ is a homeomorphism. $\square$

Definition 1.1.64. *For a G -space X , the orbit space is X/G .*

Proposition 1.1.65. *If G is compact and X is a Hausdorff G -space, then X/G is Hausdorff.*

Proof. ³ First consider the equivalence relation given on X where xRy if x and y lie in the same orbit. This equivalence relation is then given by the following set:

$$R := \{(x, gx) : x \in X, g \in G\} = \bigcup_{x \in X} \{x\} \times Gx \subseteq X \times X.$$

We wish to show that R is closed, so let $\{(x_a, g_a x_a)\}_{a \in A} \subseteq R$ be a convergent net with $(x_a, g_a x_a) \rightarrow (x, y) \in X \times X$. Since G is compact, the net $\{g_a\}_{a \in A}$ has a convergent subnet $\{g_b\}_{b \in B}$ with $g_b \rightarrow g \in G$. Then by continuity we have that the subnet $\{(x_b, g_b x_b)\}_{b \in B}$ converges to $(x, gx) \in R$ so $y = gx$. Thus R is closed.

Additionally, observe that $x \mapsto gx$ is a homeomorphism for any $g \in G$, so for any open $U \subseteq X$, $GU = \bigcup_{g \in G} gU$ is open as well. Then with the quotient map $q : X \rightarrow X/G$, $q^{-1}[GU] = GU$ since GU is a union of orbits, so $GU \in X/G$ is open.

Now let $Gx, Gy \in X/G$ with $Gx \neq Gy$, so clearly x and y do not lie in the same orbit and $(x, y) \notin R$. Because R is closed in $X \times X$, we can find open sets $U, V \subseteq X$ such that $(x, y) \in U \times V$ and $(U \times V) \cap R = \emptyset$. We immediately have that $GU \cap GV = \emptyset$ as well since otherwise x and y would be in the same orbit. Therefore we conclude that X/G is Hausdorff. $\square$

Definition 1.1.66. *Given two G -spaces X and Y , $f : X \rightarrow Y$ is a G -map if for all $g \in G$, $f(gx) = g \cdot f(x)$. We write $f : X \rightarrow_G Y$.*

Definition 1.1.67. *Given two G -spaces X and Y , $h : X \times [0, 1] \rightarrow Y$ is a G -homotopy if h is a homotopy and $h(-, t)$ is a G -map for all $t \in [0, 1]$. Equivalently, h is a G -map with G acting trivially on $[0, 1]$ and diagonally on $X \times [0, 1]$. We write $h : X \times [0, 1] \rightarrow_G Y$ or $h_t : X \rightarrow_G Y$.*

Definition 1.1.68. *Two G -maps $f, g : X \rightarrow_G Y$ are G -homotopic if there is a G -homotopy $h_t : X \rightarrow_G Y$ such that $h_0 = f$ and $h_1 = g$. We write $f \simeq_G g$.*

Definition 1.1.69. *Two G -spaces X and Y are G -homotopy equivalent if there are G -maps $f : X \rightarrow_G Y$ and $g : Y \rightarrow_G X$ such that $g \circ f \simeq_G \text{id}_X$ and $f \circ g \simeq_G \text{id}_Y$.*

Definition 1.1.70. *A set $U \subseteq X$ is invariant if $gU \subseteq U$ for all $g \in G$. Note that $U \subseteq GU$ always, so U is invariant if and only if $U = GU$.*

Definition 1.1.71. *A G -map $f : X \rightarrow Y$ is a fixed orbit map if $f[X]$ is contained in a single orbit in Y .*

Definition 1.1.72. *An invariant set $U \subseteq X$ is G -categorical if the inclusion $i : U \rightarrow X$ is G -homotopic to a fixed orbit map.*

Definition 1.1.73. *The equivariant category of a G -space X , denoted $\text{cat}_G(X)$, is the least integer k such that X can be covered by k G -categorical open sets.*

Definition 1.1.74. *A G -space X is G -contractible if $\text{cat}_G(X) = 1$.*

³Got proof idea from this conversation with Claude.

Proposition 1.1.75. *X is G -contractible if and only if X is G -homotopy equivalent to G .*

Proof. First assume that X is G -contractible, and let $h_t : X \rightarrow_G X$ be a corresponding G -homotopy witnessing this where $h_0 = \text{id}_X$ and h_1 is a fixed orbit map. Let $x_0 \in X$ be some representative of the orbit $h_1[X]$, i.e. $Gx_0 = h_1[X]$. Observe that Gx_0 is trivially homeomorphic to G via the map $gx_0 \mapsto g$, so we conclude that $X \simeq_G G$.

Conversely, assume that $X \simeq_G G$ via $f : X \rightarrow_G G$ and $g : G \rightarrow_G X$ with corresponding G -homotopy $h_t : X \rightarrow_G X$ where $h_0 = g \circ f$ and $h_1 = \text{id}_X$. Observe that g must be a fixed orbit map since $g(h \cdot e) = h \cdot g(e)$ so $g[G] = Gg(e)$. Then $g \circ f$ is trivially a fixed orbit map since $(g \circ f)[X] \subseteq g[G]$, so we conclude that X is G -contractible via h_t . $\square$

Corollary 1.1.76. *For any topological group G acting on itself in the usual way, we have that $\text{cat}_G(G) = 1$.*

Proof. Clearly $G \simeq_G G$, so by proposition 1.1.75 we are done. $\square$

Proposition 1.1.77. *For any G -space X , $\text{cat}_G(X) \geq \text{cat}(X/G)$.*

Proof. Take $\text{cat}_G(X)$ -many G -categorical open sets U_j that cover X . Since each U_j is invariant, $U_j = GU_j \subseteq X/G$ is open. Then $i_j : U_j \hookrightarrow X$ is G -homotopic to a fixed orbit map, so immediately we have that $q \circ i_j : GU_j \hookrightarrow X/G$ is homotopic to a constant map and each GU_j is null homotopic with $X/G = \cup_j GU_j$. $\square$

Proposition 1.1.78. *If X is metrizable and acted freely upon by G , then $\text{cat}_G(X) = \text{cat}(X/G)$.*

Definition 1.1.79. *The equivariant sectional category of a G -map $p : E \rightarrow_G B$, denoted $\text{secat}_G(p)$, is the least integer k such that B can be covered by k invariant open sets U_j that each admit a local G -homotopy section, i.e. there exists $s_j : U_j \rightarrow E$ such that $p \circ s \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion.*

Proposition 1.1.80. *If $p : E \rightarrow_G B$ is a G -fibration, then $\text{secat}_G(p) \leq k$ if and only if B can be covered by k invariant open sets that each admit a local G -section.*

Proof. This follows by an argument analogous to remark 1.1.51. $\square$

Definition 1.1.81. *Given a G -space X and a closed subgroup $H \subseteq G$, the H -fixed point set of X is*

$$X^H := \{x \in X : \forall h \in H, hx = x\} = \{x \in X : Hx = \{x\}\}.$$

Definition 1.1.82. *A G -space X is G -connected if X^H is path-connected for every closed subgroup $H \subseteq G$.*

Proposition 1.1.83. *Let $p : E \rightarrow B$ be a G -map where B is G -connected and $E^G \neq \emptyset$. Then $\text{secat}_G(p) \leq \text{cat}_G(B)$.*

Proposition 1.1.84. *For a G -map $p : E \rightarrow_G B$, if $p[E^H] = B^H$ for all closed subgroups $H \subseteq G$, then $\text{secat}_G(p) \leq \text{cat}_G(B)$.*

Proposition 1.1.85. *If E is G -contractible then for any G -map $p : E \rightarrow_G B$ we have $\text{cat}_G(B) \leq \text{secat}_G(p)$.*

Proof. Since E is G -contractible, we have that $\text{id}_E \simeq_G c$ where $c : E \rightarrow_G E$ is a fixed orbit map. Now take $\text{secat}_G(p)$ -many invariant open sets $U_j \subseteq B$ such that $B = \cup_j U_j$ with maps $s_j : U_j \rightarrow E$ such that $s_j \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion. Then we have

$$i_j \simeq_G p \circ s_j = p \circ \text{id}_E \circ s_j \simeq_G p \circ c \circ s_j$$

in which the latter map is a fixed orbit map since c is a fixed orbit map. Hence we conclude $\text{cat}_G(B) \leq \text{secat}_G(p)$. For clarity, this is represented in the following commutative diagram:

$$\begin{array}{ccc} E & \xrightleftharpoons[\text{c}]{\text{id}_E} & E \\ s_j \uparrow & \searrow p & \downarrow p \\ U_j & \xrightarrow{i_j} & B \end{array}$$

□

Corollary 1.1.86. *Let $p : E \rightarrow_G B$ be a G -map and let E be a G -contractible space such that one of the following holds:*

1. B is G -connected and $E^G \neq \emptyset$
2. $p[E^H] = B^H$ for all closed subgroups $H \subseteq G$.

Then $\text{secat}_G(p) = \text{cat}_G(p)$.

Proof. This immediately follows from propositions 1.1.83, 1.1.84, and 1.1.85. □

1.1.3 Background from [DLS24]

Definition 1.1.87. A map $f : X \rightarrow Y$ is (G) -invariant if for all $x \in X$ and for all $g \in G$, $f(g \cdot v) = f(v)$.

Definition 1.1.88. For a space X and group G , we denote X as a G -space with G acting trivially by X_{triv} .

Definition 1.1.89. We denote the category of topological spaces as **Top** and the category of G -spaces for fixed group G as G -**Top**.

Remark 1.1.90. A map $f : X \rightarrow Y$ is invariant if and only if it is a G -map when G acts trivially on Y , i.e. $f : X \rightarrow_G Y_{triv}$.

Remark 1.1.91. An invariant map $f : X \rightarrow Y$ induces a map $\hat{f} : X/G \rightarrow Y$ via $\hat{f}(Gv) = f(v)$. Conversely, a map $\hat{f} : X/G \rightarrow Y$ induces an invariant map $f : X \rightarrow Y$ via $f(v) = \hat{f}(Gv)$. This gives an isomorphism $\mathbf{Top}(X/G, Y) \cong G\text{-}\mathbf{Top}(X, Y_{triv})$.

Furthermore, let $L : G\text{-}\mathbf{Top} \rightarrow \mathbf{Top}$ and $R : \mathbf{Top} \rightarrow G\text{-}\mathbf{Top}$ be the functors defined as follows:

- $LX = X/G$, $L(f : X \rightarrow_G Y) = (Gx \mapsto G(f(x)))$
- $RY = Y_{triv}$, $R(f : Y \rightarrow X) = (y \mapsto f(y))$.

Then the previous isomorphism actually means that L is a left adjoint of R .

Definition 1.1.92. Given a G -space X , an orbit canonicalization is an invariant map $f : X \rightarrow X$ such that $f(x) \in Gx$ for all $x \in X$.

Remark 1.1.93. An orbit canonicalization $f : X \rightarrow X$ must have a unique fixed point in each orbit $Gx \in X/G$.

Proof. Remark 1.1.91 gives us an embedding $X/G \hookrightarrow X$, and we conclude by instead considering this embedding as a map $X \rightarrow X$. $\square$

Proposition 1.1.94. Every orbit canonicalization $f : X \rightarrow X$ corresponds uniquely to a continuous section $s : X/G \rightarrow X$.

Proof. First, let $f : X \rightarrow X$ be an orbit canonicalization. As per remark 1.1.91, f induces $\hat{f} : X/G \rightarrow X$ via $\hat{f}(Gx) = f(x)$ with $\hat{f} \circ q = f$. Since f is invariant, we have that $q(\hat{f}(Gx)) = q(f(x)) = G(f(x)) = Gx$ and so $\hat{f}$ is a section of q .

Conversely, let $s : X/G \rightarrow X$ be some section of q such that $s(Gx) \in Gx$ always. Then we can define a map $f : X \rightarrow X$ via $f = s \circ q$ and observe that $f(x) = s(q(x)) = s(Gx)$. Since s is a section of q , we have that $q(s(Gx)) = Gx$, meaning that $f(x) = s(Gx) \in q^{-1}[Gx] = Gx$. Therefore f is an orbit canonicalization.

Lastly, to see the uniqueness, observe that the process in the converse step applied to $\hat{f}$ yields f , so the map $f \mapsto \hat{f}$ is a bijection. $\square$

Corollary 1.1.95. An orbit canonicalization $f : X \rightarrow X$ exists if and only if there is a continuous section $s : X/G \rightarrow X$.

Proof. This is just a restatement of proposition 1.1.94. $\square$

1.1.4 Stiefel-Whitney Classes

The following material is from [Hat09]

Definition 1.1.96. Let E and B be spaces and let $p : E \rightarrow B$ be a map such that

1. p is surjective
2. for all $b \in B$, $p^{-1}[\{b\}]$ has an $\mathbb{R}$ -vector space structure,
3. and there is a cover of B by open sets U_α with homeomorphisms $\Phi_\alpha : p^{-1}[U_\alpha] \rightarrow U_\alpha \times \mathbb{R}^n$ such that $\Phi_\alpha|_{p^{-1}[\{b\}]} : p^{-1}[\{b\}] \rightarrow \{b\} \times \mathbb{R}^n$ is a linear isomorphism for each $b \in U_\alpha$.

Then p is an n -dimensional vector bundle and the maps Φ_α are local trivializations.

Remark 1.1.97. An n -dimensional vector bundle is a special type of fiber bundle with fiber $\mathbb{R}^n$.

Proposition 1.1.98. A vector bundle $p : E \rightarrow B$ is trivial (i.e. isomorphic to the product $B \times \mathbb{R}^n$) if and only there are n linearly independent sections $s_1, \dots, s_k : B \rightarrow E$ of p .

Definition 1.1.99. Let $p : E \rightarrow B$ be a vector bundle and let $f : B' \rightarrow B$ be some map. Then we can define the pullback bundle induced by f to be

$$E' = f^*(E) := \{(b', e) \in B' \times E : f(b') = p(e)\}.$$

Remark 1.1.100. From definition 1.1.99, the map $p' : E' \rightarrow B'$ given by $p'(b', e) = b'$ is also a vector bundle. Additionally, E' is the unique pullback $B' \times_B E$ in the category **Top**.

Definition 1.1.101. The Grassmann manifold $\text{Gr}(n, k)$ is the space of all k -dimensional vector subspaces of $\mathbb{R}^n$.

Remark 1.1.102. There is a natural vector bundle over $\text{Gr}(n, k)$ given by

$$E(n, k) := \{(P, x) \in \text{Gr}(n, k) \times \mathbb{R}^n : x \in P\}$$

with the projection $p : E(n, k) \rightarrow \text{Gr}(n, k)$.

Definition 1.1.103. The Stiefel manifold $\text{St}(n, k)$ is the space of all orthonormal bases for k -dimensional vector subspaces of $\mathbb{R}^n$.

Remark 1.1.104. There is a natural vector bundle over $\text{St}(n, k)$ given by

$$F(n, k) := \{(B, x) \in \text{St}(n, k) \times \mathbb{R}^n : x \in \text{span } B\}$$

with the projection $q : F(n, k) \rightarrow \text{Gr}(n, k)$. This bundle is trivial by proposition 1.1.98 since one clearly has k orthogonal (and hence linearly independent) sections of q given by $s_i : \text{St}(n, k) \rightarrow F(n, k)$, $s_i(B) = (B, b_i)$ where we write B in the standard basis of $\mathbb{R}^n$ with $B = [b_1 \cdots b_k]$.

Remark 1.1.105. There is a natural projection $\pi : \text{St}(n, k) \rightarrow \text{Gr}(n, k)$ given by $\pi(B) = \text{span } B$.

Theorem 1.1.106. (Theorem 3.1 in [Hat09]) There exist unique functions, the Stiefel-Whitney classes, $w_1, w_2, \dots$ that map each real vector bundle $E \rightarrow B$ to a class $w_i(E) \in H^i(B; \mathbb{Z}_2)$ such that

1. If $E \cong E'$ as vector bundles, $w_i(E) = w_i(E')$.
2. For a function $f : A \rightarrow B$ with induced map on homology $f^* : H^*(B; \mathbb{Z}_2) \rightarrow H^*(A; \mathbb{Z}_2)$ and pullback $f^*(E)$, we have $w_i(f^*(E)) = f^*(w_i(E))$.
3. $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \dots \in H^*(B; \mathbb{Z}_2)$.
4. $w_i(E) = 0$ if $i > \dim E$.
5. For the canonical line bundle $E \rightarrow \mathbb{R}P^\infty$, $w_1(E)$ is a generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2)$.

Remark 1.1.107. A trivial vector bundle has $w_i(E) = 0$ for $i > 0$.

Proposition 1.1.108. (From [Jaw06]) Let $w_1, \dots, w_k$ be the Stiefel-Whitney classes of the bundle $p : E(n, k) \rightarrow \text{Gr}(n, k)$, let $\bar{w}_1, \dots, \bar{w}_n$ be the dual classes, and let $J(n, k)$ be the ideal in $\mathbb{Z}_2[w_1, \dots, w_k]$ generated by the relation $w\bar{w} = 1$. Then

$$H^*(\text{Gr}(n, k)) \cong \mathbb{Z}_2[w_1, \dots, w_k] / J(n, k).$$

Proposition 1.1.109. Let $\pi : \text{St}(n, k) \rightarrow \text{Gr}(n, k)$ be the natural projection. Then the induced map $\pi^* : H^*(\text{Gr}(n, k)) \rightarrow H^*(\text{St}(n, k))$ is zero on positive-dimensional cohomology.

Proof. Let $\pi^*(E(n, k))$ be the pullback of the vector bundle $p : E(n, k) \rightarrow \text{Gr}(n, k)$ via $\pi : \text{St}(n, k) \rightarrow \text{Gr}(n, k)$. Then observe that

$$\begin{aligned} \pi^*(E(n, k)) &= \{(B, (P, x)) \in \text{St}(n, k) \times E(n, k) : \pi(B) = p(P, x) = P\} \\ &= \{(B, x) \in \text{St}(n, k) \times \mathbb{R}^n : x \in \text{span } B\} \\ &= F(n, k). \end{aligned}$$

Thus the pullback bundle is actually just the bundle $q : F(n, k) \rightarrow \text{St}(n, k)$. Then by theorem 1.1.106 we have that $\pi^*(w_i(E(n, k))) = w_i(F(n, k)) = 0$ since $q : F(n, k) \rightarrow \text{St}(n, k)$ is a trivial bundle. By proposition 1.1.108 we know that $H^*(\text{Gr}(n, k); \mathbb{Z}_2)$ is generated by (a quotient of) the Stiefel-Whitney classes $w_i(E(n, k))$, so we conclude that $\pi^* = 0$ on positive-dimensional cohomology. $\square$

Corollary 1.1.110. $\text{nil ker } \pi^* = \text{cup}(\text{Gr}(n, k)) - 1$

1.2 New Work

1.2.1 Sectional Category and Canonicalization

Proposition 1.2.1. *If there exists an orbit canonicalization $f : X \rightarrow X$, then $\text{secat}(q) = 1$ where $q : X \rightarrow X/G$ is the quotient map.*

Proof. By proposition 1.1.94 we know that f induces a section $\hat{f} : X/G \rightarrow X$ of q , so we conclude that $\text{secat}(q) = 1$. $\square$

Corollary 1.2.2. *If $\text{secat}(q) > 1$, there is no orbit canonicalization $f : X \rightarrow X$.*

With this in mind, we now wish to study the sectional category of the quotient map $q : X \rightarrow X/G$.

Definition 1.2.3. *For spaces E and B , a map $p : E \rightarrow B$ has the homotopy lifting property up to homotopy (HLPH) for a space X if for every homotopy $h_t : X \rightarrow B$ and every homotopy lift $\tilde{h}_0 : X \rightarrow E$ of h_0 ($h_0 \simeq p \circ \tilde{h}_0$), there exists a homotopy $\tilde{h}_t : X \rightarrow E$ that lifts h_t up to homotopy ($h_t \simeq p \circ \tilde{h}_t$).*

Proposition 1.2.4. *For any spaces E and B and a surjective map $p : E \rightarrow B$, $\text{secat}(p) \leq \text{cat}(B)$ as in definitions 1.1.13 and 1.1.16.*

Proof. Take $\text{cat}(B)$ -many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \rightarrow B$. $\square$

Definition 1.2.5. *For spaces E and B , a map $p : E \rightarrow B$ is a fibration up to homotopy if p satisfies the HLPH for all topological spaces.*

Remark 1.2.6. *Clearly HLP implies HLPH and any fibration is a fibration up to homotopy.*

Lemma 1.2.7. *Let X be a G -space and let $q : X \rightarrow X/G$ be the quotient map. Then if $U \subseteq X$ is G -categorical, the set $q[U] \subseteq X/G$ is contractible.*

Proof. First, let $U \subseteq X$ be G -categorical, i.e. the inclusion $i : U \hookrightarrow X$ is G -homotopic to a fixed orbit map $c : U \rightarrow X$. This clearly induces a homotopy from the inclusion $q \circ i : q[U] \hookrightarrow X/G$ to $q \circ c : q[U] \rightarrow X/G$, and since c is a fixed orbit map $q \circ c$ must be a constant map. Thus $q[U]$ is contractible. $\square$

Lemma 1.2.8. *Let X be a G space such that the quotient map $q : X \rightarrow X/G$ is a fibration. Then if $U \subseteq X/G$ is open and contractible, the set $q^{-1}[U] \subseteq X$ is G -categorical.*

Proof. Let $h_t : U \rightarrow X/G$ be a homotopy witnessing that U is contractible, with h_0 being the inclusion $i : U \hookrightarrow X/G$ and h_1 being a constant map $c : U \rightarrow X/G$. Set $V = q^{-1}[U]$, noting that V is open and invariant, and let $\tilde{h}_0 : V \rightarrow X$ be the inclusion $\tilde{i} : V \hookrightarrow X$. Then clearly $q \circ \tilde{h}_0 = i$, so by the homotopy lifting property we have a homotopy $\tilde{h}_t : V \rightarrow X$ such that $q \circ \tilde{h}_t = h_t$. Specifically this means $q \circ \tilde{h}_1 = c$, i.e. $\tilde{h}_1 : V \rightarrow X$ is a fixed orbit map, so we conclude that V is G -categorical. $\square$

Proposition 1.2.9. *If X is a G -space such that the quotient map $q : X \rightarrow X/G$ is a fibration up to homotopy, then $\text{secat}(q) \leq \text{cat}_G(X)$.*

Proof. Take $\text{cat}_G(X)$ -many G -categorical sets $U_j \subseteq X$ such that $X = \cup_j U_j$. Define $V_j = q[U_j]$, and note that by lemma 1.2.7 each V_j is contractible. Fix some j and let $h_t : V_j \rightarrow X/G$ be a homotopy witnessing the contractibility of V_j , with h_0 being a constant map $c_j : V_j \rightarrow X/G$ and h_1 being the inclusion $i_j : V_j \hookrightarrow X/G$. Then fix some $x_0 \in q^{-1}[c_j[V_j]]$ and define $\tilde{h}_0 : V_j \rightarrow X$ by $\tilde{h}_0 = c_{x_0}$. Clearly $q \circ \tilde{h}_0 = c_j$, so the HLPH gives us a homotopy $\tilde{h}_t : V_j \rightarrow X$ such that $q \circ \tilde{h}_t \simeq h_t$. In particular, this means $q \circ \tilde{h}_1 \simeq i_j$ so $\tilde{h}_1$ is a homotopy section of q . Repeating this for all j , we conclude that $\text{secat}(q) \leq \text{cat}_G(X)$. $\square$

Definition 1.2.10. *The contractible sectional category of a map $p : E \rightarrow B$, denoted by $\text{consecat}(p)$, is the least integer k such that B may be covered by k contractible open sets U_j that each admit a map $s_j : U_j \rightarrow E$ where $p \circ s_j$ is homotopic to the inclusion $i_j : U_j \hookrightarrow B$.*

Remark 1.2.11. *Trivially, $\text{consecat}(p) \geq \text{secat}(p)$ for any map p .*

Proposition 1.2.12. *If X is a G -space such that the quotient map $q : X \rightarrow X/G$ is a fibration up to homotopy, then $\text{cat}_G(X) \leq \text{consecat}(q)$.*

Proof. Take $\text{consecat}(q)$ -many contractible open sets $U_j \subseteq X/G$ with $X/G = \cup_j U_j$ that each admit a homotopy section $s_j : U_j \rightarrow X$ of q . Define $V_j = q^{-1}[U_j]$, and note that by lemma 1.2.8 that each V_j is G -categorical. Clearly $X = \cup_j V_j$, so we conclude that $\text{cat}_G(X) \leq \text{consecat}(q)$. $\square$

Question 1.2.13. *The homotopy sections in the proof above are completely unused. Can something stronger be proven?*

Question 1.2.14. *When do we have $\text{consecat}(p) = \text{secat}(p)$? Or are there any other bounds we can put on $\text{secat}(p)$ via $\text{consecat}(p)$?*

Recall that when $p : E \rightarrow B$ is a fibration we have that definitions 1.1.13 and 1.1.50 coincide (i.e. homotopy sections induce sections), so we stand to learn a lot about orbit canonicalizations if we can discern when $q : X \rightarrow X/G$ is a fibration.

Theorem 1.2.15. *This is part of theorem 1.8 in [Rai06]. A fiber bundle $p : E \rightarrow B$ over paracompact space B is a fibration.*

1.2.2 Bounds for Strong and Weak Sectional Category

We first introduce some (slightly) new terminology.

Definition 1.2.16. *The weak sectional category of a map $p : E \rightarrow B$, denoted $\text{secat}(p)$, is the least integer k such that there k open sets U_j covering B that each admit a homotopy section $s_j : U_j \rightarrow E$ of p (i.e. $p \circ s_j \simeq i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Definition 1.2.17. *The strong sectional category of a map $p : E \rightarrow B$, denoted $\text{secat}^s(p)$, is the least integer k such that there k open sets U_j covering B that each admit a section $s_j : U_j \rightarrow E$ of p (i.e. $p \circ s_j = i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Recall that by remark 1.1.51 $\text{secat}(p) = \text{secat}^s(p)$ when p is a fibration. In general, we have the following result:

Lemma 1.2.18. *For any map $p : E \rightarrow B$, $\text{secat}(p) \leq \text{secat}^s(p)$.*

Proof. Take $\text{secat}^s(p)$ -many open sets covering U_j with sections $s_j : U_j \rightarrow E$ of p i.e. $p \circ s_j = i_j$. It trivially follows that $p \circ s_j \simeq i_j$, so we are done. $\square$

Definition 1.2.19. *(Definition 1.1 in [Coh26]) Let E , B , and F be spaces, and let $p : E \rightarrow B$ be a map such that*

1. *p is surjective,*
2. *for all $b \in B$, $p^{-1}[\{b\}] \cong F$,*
3. *and for all $b \in B$, there is some open neighborhood $U_b \subseteq B$ of b and a homeomorphism $\Phi_b : p^{-1}[U_b] \rightarrow U_b \times F$ such that $p|_{p^{-1}[U_b]} = \pi_b \circ \Phi_b$ where $\pi_b : U_b \times F \rightarrow U_b$ is the projection.*

Then p is a fiber bundle with fiber F .

Lemma 1.2.20. *(Corollary 2.7.14 in [Spa66]) If B is paracompact, then any fiber bundle $p : E \rightarrow B$ is a fibration.*

Definition 1.2.21. *(Definition 1.5 in [Coh26]) A topological group G is a Lie group is G is a differentiable manifold such that products and inverses are differentiable maps.*

Definition 1.2.22. *(Definition 1.6 in [Coh26]) Let G be a topological group, let E and B be spaces, and let $p : E \rightarrow B$ be a fiber bundle with fiber G such that*

1. *E has a free left G -action where $p(e) = p(ge)$ for all $g \in G$, $e \in E$, i.e. the fibers of p are closed under the G -action,*
2. *for all $b \in B$, the induced G -action on $p^{-1}[\{b\}]$ is free and transitive,*
3. *and for every open set $U \subseteq B$, there exists a G -equivariant homeomorphism $\Psi_U : p^{-1}[U] \rightarrow G \times U$ where G acts on $G \times U$ by multiplication on G .*

Then p is a principal G -bundle.

Definition 1.2.23. (Definition 1.7 in [Coh26]) A G -space X has slices if the quotient map $q : X \rightarrow X/G$ has local sections around every point in X/G .

Proposition 1.2.24. (Proposition 1.2 in [Coh26]) If X is a free G -space with slices, then the quotient map $q : X \rightarrow X/G$ is a principal G -bundle.

Theorem 1.2.25. (Theorem 1.3 in [Coh26]) Let M be a smooth manifold with a smooth, free G -action by a compact Lie group G . Then M has slices.

Corollary 1.2.26. Let M be a smooth manifold with a smooth, free G -action by a compact Lie group G . Then $q : M \rightarrow M/G$ is a principal G -bundle.

Proof. This follows directly from theorem 1.2.25 and proposition 1.2.24. $\square$

Theorem 1.2.27. Let M be a smooth manifold with a smooth, free G -action by a compact Lie group G such that M/G is paracompact. Then $q : M \rightarrow M/G$ is a fibration.

Proof. This follows directly from corollary 1.2.26 and lemma 1.2.20. $\square$

Theorem 1.2.28. Let G be a compact Lie group and let M be a smooth manifold with submanifold $N \subseteq M$ such that

1. G acts smoothly on M ,
2. N is G -invariant (hence $N/G \subseteq M/G$),
3. the induced G -action on N is free,
4. and N/G is paracompact.

Then letting $q : M \rightarrow M/G$ and $p = q|_N : N \rightarrow N/G$ be the respective quotient maps and letting $p^* : H^*(N/G) \rightarrow H^*(N)$ be the induced map on cohomology, we have the following chain of inequalities:

$$\text{nil ker}(p^*) < \text{secat}(p) = \text{secat}^s(p) \leq \text{secat}^s(q).$$

Proof. Since G is a compact Lie group acting freely on N and N/G is paracompact, by theorem 1.2.27 we know that p is a fibration. Then by proposition 1.1.58 we have that $\text{nil ker}(p^*) < \text{secat}(p)$; by remark 1.1.51 we have that $\text{secat}(p) = \text{secat}^s(p)$; and by lemma 1.1.19 we have that $\text{secat}^s(p) \leq \text{secat}^s(q)$. $\square$

Theorem 1.2.28 gives a concrete method to find a lower bound for strong sectional category even for quotient maps that fail to be fibrations by considering appropriate submanifolds of a given space. This is particularly useful in the problem of orbit canonicalization, since orbit canonicalizations are directly related to the strong sectional category of the quotient map.

1.2.3 Canonicalizing $\mathbb{R}^{d \times n}$ with respect to $O(d)$ and $SO(d)$

Definition 1.2.29. The orthogonal group of degree n is the set

$$O(n) := \{Q \in \mathbb{R}^{n \times n} : QQ^T = I\}.$$

Definition 1.2.30. The special orthogonal group of degree n is the set

$$SO(n) := \{Q \in \mathbb{R}^{n \times n} : \det(Q) = 1\}.$$

Remark 1.2.31. Observe that $SO(n) \subseteq O(n)$.

Definition 1.2.32. The set of real symmetric matrices of degree n is

$$\mathbb{S}^n := \{A \in \mathbb{R}^{n \times n} : A = A^T\}.$$

Definition 1.2.33. The set of positive semi-definite matrices of degree n is

$$\mathbb{S}_+^n := \{A \in \mathbb{S}^n : \forall x \in \mathbb{R}^n, x^T A x \geq 0\}.$$

Definition 1.2.34. The set of negative semi-definite matrices of degree n is

$$\mathbb{S}_-^n := \{A \in \mathbb{S}^n : \forall x \in \mathbb{R}^n, x^T A x \leq 0\}.$$

Definition 1.2.35. The set of positive semi-definite matrices of degree n with at most rank d is

$$\mathbb{S}_{\leq d}^n := \{A \in \mathbb{S}_+^n : \text{rank}(A) \leq d\}.$$

Remark 1.2.36. $\mathbb{S}_{\leq d}^n = \mathbb{S}_{\leq n}^n$ for any $d \geq n$.

Proposition 1.2.37. $A \in \mathbb{S}_+^n$ if and only if there is some matrix $B \in \mathbb{R}^{n \times n}$ such that $A = B^T B$ [Wik26b].

Corollary 1.2.38. $\mathbb{S}_+^n = \mathbb{S}_{\leq n}^n$.

Definition 1.2.39. The set of positive semi-definite matrices of degree n with rank d is

$$\mathbb{S}_d^n := \{A \in \mathbb{S}_+^n : \text{rank}(A) = d\}.$$

Proposition 1.2.40. $A \in \mathbb{S}_d^n$ if and only if there is some matrix $B \in \mathbb{R}^{d \times n}$ such that $A = B^T B$ with $\text{rank}(B) = d$. [Wik26b].

Lemma 1.2.41. With $O(d)$ acting on $\mathbb{R}^{d \times n}$ via matrix multiplication on the left, $\mathbb{R}^{d \times n} / O(d)$ is homeomorphic to the space $S := \{A^T A : A \in \mathbb{R}^{d \times n}\}$.

Proof. To begin, consider the map $\phi : \mathbb{R}^{d \times n}$ given by $\phi(A) = A^T A$. Now let $A, B \in \mathbb{R}^{d \times n}$ such that $[A] = [B] \in \mathbb{R}^{d \times n} / O(d)$, i.e. there exists $Q \in O(d)$ such that $A = QB$. This means that $A^T A = (QB)^T QB = B^T Q^T QB = B^T B$, so ϕ induces a map $\hat{\phi} : \mathbb{R}^{d \times n} / O(d) \rightarrow S$ via $\hat{\phi}([A]) = A^T A$. Note that $\hat{\phi}$ is bijective by construction, with inverse $\hat{\phi}^{-1} : S \rightarrow \mathbb{R}^{d \times n} / O(d)$ given by $\hat{\phi}^{-1}(A^T A) = [A]$. Both of these maps are continuous, so we are done. $\square$

Corollary 1.2.42. $\mathbb{R}^{d \times n}/O(d) \cong \mathbb{S}_{\leq d}^n$.

Proof. This follows directly from proposition 1.2.40 and lemma 1.2.41. $\square$

With corollary 1.2.42 in hand, we have that any invariant map $f : \mathbb{R}^{d \times n} \rightarrow Y$ factors through $\mathbb{S}_d^n$ and so it suffices to reduce to consider maps from $\mathbb{S}_d^n$ to Y .

Definition 1.2.43. Given a group G and a G -set X , the free locus of X is

$$X_{free} := \{x \in X : \forall g \in G \setminus \{e\}, gx \neq x\} = \{x \in X : G_x = \{e\}\}$$

Lemma 1.2.44. With $O(d)$ acting on $\mathbb{R}^{d \times n}$ via matrix multiplication, $\mathbb{R}_{free}^{d \times n} = \mathbb{R}_*^{d \times n}$.

Proof. First, let $A \in \mathbb{R}^{d \times n}$ such that $\text{rank}(A) = k < d$. Then there are some orthonormal bases $\{e_1, \dots, e_k\}$ of $\text{col}(A)$ and $\{e_{k+1}, \dots, e_d\}$ of $\text{null}(A)$. Let $B = [e_1 \cdots e_d]^T \in O(d)$ be the change of basis matrix from the standard basis for $\mathbb{R}^d$ to $\{e_1, \dots, e_d\}$. Then A can be written in the basis $\{e_1, \dots, e_d\}$ as a matrix

$$\hat{A} = BA = \begin{bmatrix} \hat{a}_{1,1} & \cdots & \hat{a}_{n,1} \\ \vdots & & \vdots \\ \hat{a}_{k,1} & \cdots & \hat{a}_{n,k} \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}.$$

Now define the matrix $R \in \mathbb{R}^{d \times d}$ by $R_{i,i} = 1$ if $i \leq k$, $R_{i,i} = -1$ if $i > k$, and $R_{i,j} = 0$ if $i \neq j$ (R simply flips the sign of the last $d - k$ coordinates) and observe that clearly $R \in O(d)$. Define $Q = B^T R B$, and then $Q \in O(d)$ since

$$Q^T Q = (B^T R B)^T B^T R B = B^T R^T B B^T R B = I.$$

By construction we have that $R\hat{A} = \hat{A}$, so it follows that $QA = A$. However, without loss of generality $Q \neq I$, so we have that $A \notin \mathbb{R}_{free}^{d \times n}$.

Conversely, let $A \in \mathbb{R}_*^{d \times n}$ and let $Q \in O(d)$ such that $QA = A$. Since $\text{col}(A) = \mathbb{R}^d$, this means that Q fixes all of $\mathbb{R}^d$ and hence $Q = I$. Therefore $A \in \mathbb{R}_{free}^{d \times n}$. $\square$

Corollary 1.2.45. $O(d)$ does not act freely on any subset of $\mathbb{R}^{d \times n}$ if $n < d$.

Corollary 1.2.46. $\mathbb{R}_*^{d \times n}/O(d) \cong \mathbb{S}_d^n$.

Proof. This follows from proposition 1.2.40, lemma 1.2.41, and lemma 1.2.44. $\square$

We now present a somewhat simplified proof of the result of [CG13] that $(\mathbb{R}^{d \times n}, O(d))$ and $(\mathbb{R}^{d \times n}, SO(d))$ lack continuous canonicalizations for $n > d$ and $n \geq d$ respectively. Propositions 1.2.49 and 1.2.50 and theorem 1.2.51 are due to Casson, as presented in theorem 1.1 of [Ken+99].

Definition 1.2.47. For a finite dimensional matrix $A \in \mathbb{R}^{n \times m}$, the spectrum of A is the set of (possibly complex) eigenvalues of A , denoted $\sigma(A)$.

Note: this is a slight abuse of the standard definition of the spectrum of a linear operator, but since we are in finite dimensions this definition coincides with the more general one.

Lemma 1.2.48. Let $E \in \mathbb{S}^n \setminus \{0\}$ with $\text{tr}(E) = 0$, and let $\varepsilon \in \mathbb{R}$ be the least eigenvalue of E . Then $|\varepsilon| > \frac{\|E\|}{n}$ where $\|\cdot\|$ denotes the operator norm.

Proof. Assume for the sake of contradiction that $\varepsilon \leq \frac{\|E\|}{n}$. Let $\{\lambda_i\}$ and $\{\mu_j\}$ be the positive and negative eigenvalues of E respectively. Recall that $\text{tr}(E) = \sum_i \lambda_i + \sum_j \mu_j$, so E must have at least one positive and one negative eigenvalue since $E \neq 0$. Also note that we must have that $|\mu_j| \leq |\varepsilon|$ for all j . From functional analysis we know that $\|E\| = \max\{|\min \sigma(E)|, |\max \sigma(E)|\}$, so since $|\varepsilon| < \|E\|$ we must have some positive eigenvalue $\lambda_j = \|E\|$. Then we have

$$\|E\| \leq \sum_i \lambda_i = \sum_j |\mu_j| \leq (n-1)|\varepsilon| < n|\varepsilon| \leq \|E\|$$

which is a contradiction. Therefore we conclude that $|\varepsilon| > \frac{\|E\|}{n}$. $\square$

Proposition 1.2.49. Let $Y_+ = \mathbb{S}_+^n \setminus \{0\}$ be the set of non-zero positive semidefinite matrices of degree n , let $Y_0 = \{A \in Y_+ : \det(A) = 0\}$, and let $\mathbb{R}_+$ be the group of positive real numbers under multiplication. If $\mathbb{R}_+$ acts on the left of Y_+ via scalar multiplication, then $Y_+/\mathbb{R}_+$ is homeomorphic to a closed ball of dimension $n(n+1)/2 - 1$. Additionally, $Y_0/\mathbb{R}_+$ is homeomorphic to the ball's boundary.

Proof. The basic idea of this proof is that we observe that Y_+ is a convex cone without its point, so we simply take a slice of it and homeomorphically map that slice to a disk. Then it is reasonably clear that such a slice is exactly the cone quotient by scaling, so we conclude there. The difficulty lies in showing the slice is actually a topological disk, the details of which lie in this proof.

Let $\langle A, B \rangle := \text{tr}(AB)$ be the Frobenius inner product on $\mathbb{S}^n$, which induces the Frobenius norm $\|\cdot\|_F$. Then let $I^\perp$ be orthogonal complement of I in $\mathbb{S}^n$, let $D := (I + I^\perp) \cap Y_+$, and let $B := \overline{B(0, 1)} \cap I^\perp$ be the closed unit ball in $I^\perp$ under the operator, which has dimension $\dim(I^\perp) = \dim(Y_+) - \dim(\text{span}\{I\}) = n(n+1)/2 - 1$. Observe that D is homeomorphic to $Y_+/\mathbb{R}_+$ via the quotient map $E \mapsto [E]$ with continuous inverse $[E] \mapsto \frac{1}{\|\text{proj}_I E\|} E$.

Define the function $\mu : \mathbb{S}^n \rightarrow \mathbb{R}$ by $\mu(A) = \min \sigma(A)$. μ is well-defined since real symmetric matrices have finitely many eigenvalues, all of which are real, and it follows from Weyl's inequality that μ is continuous. For any $A \in \mathbb{S}^n$, observe that $I + A \in Y_+$ if and only if $\mu(I + A) \geq 0$.

We now define $\Phi : B \rightarrow D$ via $\Phi(E) = I - \frac{\|E\|}{\mu(E)} E$ if $E \neq 0$ and $\Phi(0) = I$. We will show that Φ is the desired homeomorphism. Observe that $\Phi(E) \in (I + I^\perp)$ by construction, but it remains to show that $\Phi(E) \in Y_+$. This is trivially true for $E = 0$, so we consider $E \neq 0$. Note that $E \in I^\perp$ means that $\text{tr}(E) = 0$, meaning that the sum of E 's eigenvalues is 0, so

E must have at least one negative eigenvalue since $E \neq 0$. Hence, $\mu(E) < 0$. Then we see that $E \in Y_+$ since

$$\begin{aligned} \min_{\|x\|=1} x^T \Phi(E)x &= \min_{\|x\|=1} x^T \left(I - \frac{\|E\|}{\mu(E)} E \right) x \\ &= 1 - \frac{\|E\|}{\mu(E)} \min_{\|x\|=1} x^T E x \\ &= 1 - \|E\| \\ &\geq 0. \end{aligned}$$

This also shows that $\Phi(E) \in Y_0$ if and only if $\|E\| = 1$. The continuity of Φ at every point other than 0 is evident from its construction. To see Φ 's continuity at 0, first observe that

$$\frac{\mu(E)}{\|E\|} = \frac{\min_{\|x\|=1} x^T E x}{\|E\|} = \min_{\|x\|=1} x^T \frac{E}{\|E\|} x = \mu\left(\frac{E}{\|E\|}\right)$$

so

$$\Phi(E) = I - \frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E.$$

Note that $\Phi(E)$ is continuous if and only if $\frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E$ is, and since $\left\| \frac{E}{\|E\|} \right\| = 1$ we use lemma 1.2.48 to get that $\mu\left(\frac{E}{\|E\|}\right) > \frac{1}{n}$ and compute

$$\lim_{E \rightarrow 0} \left\| \frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E \right\| < \lim_{E \rightarrow 0} \|nE\| = 0$$

so $\lim_{E \rightarrow 0} \Phi(E) = I$ and Φ is continuous.

We show that Φ is indeed a homeomorphism by constructing its inverse $\Phi^{-1} : D \rightarrow B$ via $\Phi^{-1}(I + E) = -\frac{\mu(E)}{\|E\|} E$ for $E \neq 0$ and $\Phi^{-1}(I) = 0$. To see that this is indeed the inverse of Φ , we first compute

$$\begin{aligned} \Phi(\Phi^{-1}(I + E)) &= \Phi\left(-\frac{\mu(E)}{\|E\|} E\right) \\ &= I - \frac{\left\| -\frac{\mu(E)}{\|E\|} E \right\|}{\mu\left(-\frac{\mu(E)}{\|E\|} E\right)} \left(-\frac{\mu(E)}{\|E\|} E\right) \\ &= I + \frac{-\frac{\mu(E)}{\|E\|} \|E\|}{-\frac{\mu(E)}{\|E\|} \mu(E)} \frac{\mu(E)}{\|E\|} E \\ &= I + E \end{aligned}$$

when $E \neq 0$, and the computation is trivial when $E = 0$. Similarly, we have

$$\begin{aligned}
\Phi^{-1}(\Phi(E)) &= \Phi^{-1} \left(I - \frac{\|E\|}{\mu(E)} E \right) \\
&= -\mu \left(\frac{-\frac{\|E\|}{\mu(E)} E}{\left\| -\frac{\|E\|}{\mu(E)} E \right\|} \right) \left(-\frac{\|E\|}{\mu(E)} E \right) \\
&= \frac{-\frac{\|E\|}{\mu(E)}}{-\frac{\|E\|}{\mu(E)} \|E\|} \mu(E) \frac{\|E\|}{\mu(E)} E \\
&= E.
\end{aligned}$$

Thus we have that Φ^{-1} is indeed the inverse of Φ . Lastly, Φ^{-1} is again clearly continuous everywhere except 0 by construction, and at 0 we have by the continuity of μ that

$$\lim_{E \rightarrow 0} \|\Phi^{-1}(E)\| = \lim_{E \rightarrow 0} \left\| -\frac{\mu(E)}{\|E\|} E \right\| = \lim_{E \rightarrow 0} |\mu(E)| \frac{\|E\|}{\|E\|} = 0.$$

We conclude that Φ is a homeomorphism, hence $Y_+/\mathbb{R}_+ \cong B$ with $Y_0/\mathbb{R}_+ \cong \partial B$. $\square$

Alternative proof. First, we wish to show that the identity matrix I is interior point of $Y_+ \subseteq \mathbb{S}^n$ in the subspace topology. Recall all norms are equivalent on a finite dimensional vector space, so it suffices to find an open ball under the operator norm. Define the function $\mu : \mathbb{S}^n \rightarrow \mathbb{R}$ by $\mu(A) = \min \sigma(A)$. μ is well-defined since real symmetric matrices have finitely many eigenvalues, all of which are real, and it follows from Weyl's inequality that μ is continuous. For any $A \in \mathbb{S}^n$, observe that $I + A \in Y_+$ if and only if $\mu(I + A) \geq 0$. By the continuity of μ , there is some δ such that $\|A\| < \delta$ implies $|\mu(I) - \mu(I + A)| < 1$, meaning $\mu(I + A) > 0$ since $\mu(I) = 1$. Therefore $B(I, \delta) \subseteq Y_+$, as desired, so $\dim Y_+ = \dim \mathbb{S}^n = n(n+1)/2$.

Next, Y_+ is clearly convex since the sum of any two positive semidefinite matrices is also positive semidefinite. From here, one can simply consider the intersection $Y_+ \cap (I + I^\perp)$ which is a bounded, convex subset of Y_+ and is hence homeomorphic to the ball of dimension $n(n+1)/2 - 1$. Additionally, $\partial Y_+ = Y_0$ since any singular matrix can be perturbed to have a negative eigenvalue, so $Y_0 \cap (I + I^\perp)$ is homeomorphic to the ball's boundary. $\square$

Proposition 1.2.50. *Let $X_+ = \{A \in \mathbb{R}^{n \times n} \setminus \{0\} : \det(A) \geq 0\}$ with left $\mathrm{SO}(n)$ action given by matrix multiplication, and let $Y_+ = \mathbb{S}_+^n \setminus \{0\}$. Additionally, let $X_0 \subseteq X_+$ and $Y_0 \subseteq Y_+$ be the subspaces of singular matrices. Then $X_+/\mathrm{SO}(n) \cong Y_+$ with $X_0/\mathrm{SO}(n) \cong Y_0$.*

Proof. This follows from a nearly identical argument to that of lemma 1.2.41 and corollary 1.2.42, simply with the added care to only use matrices with nonnegative determinant. $\square$

Theorem 1.2.51 (Casson). *Let $X = \mathbb{R}^{n \times n} \setminus \{0\}$ with $\mathrm{SO}(n)$ and $\mathbb{R}_+$ acting on the left via matrix and scalar multiplication respectively, and let S be the sphere of dimension $n(n+1)/2 - 1$. Then $X/(\mathbb{R}_+ \times \mathrm{SO}(n)) \cong S$.*

Proof. First observe that the actions of $\mathbb{R}_+$ and $\mathrm{SO}(n)$ on X commute, so we have that $X/(\mathbb{R}_+ \times \mathrm{SO}(n)) \cong (X/\mathrm{SO}(n))/\mathbb{R}_+$. Let X_0, X_+, X_- be the subspaces of X of matrices with zero, nonnegative, and nonpositive determinant respectively, and observe that clearly $X \cong X_+ \cup_{X_0} X_-$. Observe that any $\mathrm{SO}(n)$ orbit is entirely contained in X_0, X_+ , or X_- , since $\mathrm{SO}(n)$ -actions do not change the determinant. Additionally, $X_+ \cong X_-$ via your favorite map that flips the sign of the determinant (e.g. multiplying the last column by -1). It then follows that

$$(X/\mathrm{SO}(n))/\mathbb{R}_+ \cong (X_+/\mathrm{SO}(n))/\mathbb{R}_+ \cup_{(X_0/\mathrm{SO}(n))/\mathbb{R}_+} (X_-/\mathrm{SO}(n))/\mathbb{R}_+.$$

By propositions 1.2.49 and 1.2.50 we know that $(X_+/\mathrm{SO}(n))/\mathbb{R}_+$ and $(X_0/\mathrm{SO}(n))/\mathbb{R}_+$ are homeomorphic to the ball of dimension $n(n+1)/2 - 1$ and its boundary respectively, so we conclude that $X/(\mathbb{R}_+ \times \mathrm{SO}(n))$ is homeomorphic to the sphere of dimension $n(n+1)/2 - 1$. $\square$

With Casson's theorem in hand we can now rather quickly prove two of [DLS24]'s impossibility results.

Lemma 1.2.52. *(From [DLS24]) Let $G \subseteq \mathbb{R}^{d \times d}$ be a group acting on $\mathbb{R}^{d \times n}$ on the left via matrix multiplication. Then a canonicalization for $\mathbb{R}^{d \times n}$ induces a canonicalization for $\mathbb{R}^{d \times m}$ for any $m \leq n$.*

Proof. Let $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ be a canonicalization, and consider the map $\hat{f} : \mathbb{R}^{d \times m} \rightarrow \mathbb{R}^{d \times m}$ given by $\hat{f}(A) = f([A \mid 0])[:, : m]$ ($[A \mid 0]$ indicates padding A on the right with zeroes so $[A \mid 0] \in \mathbb{R}^{d \times n}$). Then for any $Q \in G$ and $A \in \mathbb{R}^{d \times m}$,

$$\hat{f}(QA) = f(Q[A \mid 0])[:, : m] = (\hat{Q}[A \mid 0])[:, : m] = \hat{Q}A$$

for some $\hat{Q} \in G$. Note that $\hat{f}(QB) = \hat{Q}A$ any $Q \in G$, and $B \in GA$ since f is a canonicalization, so we conclude that $\hat{f}$ is a canonicalization as well. $\square$

Theorem 1.2.53. *If $n \geq d$, $\mathbb{R}^{d \times n}$ has no canonicalization with respect to $\mathrm{SO}(d)$.*

Proof. By lemma 1.2.52 we know that any canonicalization of $\mathbb{R}^{d \times n}$ induces a canonicalization for $\mathbb{R}^{d \times d}$, so we reduce to considering this case. Now, assume for the sake of contradiction that we have a canonicalization for $\mathbb{R}^{d \times d}$, and recall that this induces a section of the quotient map $\mathbb{R}^{d \times d} \rightarrow \mathbb{R}^{d \times d}/\mathrm{SO}(d)$. Observe that such a section also induces a section on any $\mathrm{SO}(d)$ -invariant subspace of $\mathbb{R}^{d \times d}$, so we specifically consider the unit sphere $S^{d^2-1} \subseteq \mathbb{R}^{d \times d}$ under the Frobenius norm. It is easy to see that S^{d^2-1} is invariant under $\mathrm{SO}(d)$ transformations since orthogonal transformations do not change the Frobenius norm, and it is also clear that S^{d^2-1} is homeomorphic to the quotient $(\mathbb{R}^{d \times d} \setminus \{0\})/\mathbb{R}_+$ since one can always scale a non-zero matrix to have norm 1. Hence, we have a section $s : S^{d^2-1}/\mathrm{SO}(d) \rightarrow S^{d^2-1}$ of the quotient map $q : S^{d^2-1} \rightarrow S^{d^2-1}/\mathrm{SO}(d)$.

Recall by theorem 1.2.51 that the quotient $S^{d^2-1}/\mathrm{SO}(d)$ is homeomorphic to the sphere of dimension $d(d+1)/2 - 1$, $S^{\frac{d(d+1)}{2}-1}$. So, up to slight abuse of notation, we have induced maps on homology

$$H_k(S^{\frac{d(d+1)}{2}-1}) \xrightarrow{s_\#} H_k(S^{d^2-1}) \xrightarrow{q_\#} H_k(S^{\frac{d(d+1)}{2}-1})$$

such that $q_{\sharp} \circ s_{\sharp}$ is still the identity. However, when $k = d(d+1)/2 - 1$ this is impossible, since in this case $H_k(S^{\frac{d(d+1)}{2}-1}) \cong \mathbb{Z}$ and $H_k(S^{d^2-1}) \cong 0$. Therefore we conclude that there can be no canonicalization of $\mathbb{R}^{d \times n}$ with respect to $SO(d)$ for $n \geq d$. $\square$

Theorem 1.2.54. *If $n > d$, $\mathbb{R}^{d \times n}$ has no canonicalization with respect to $O(d)$.*

Proof 1. This proof proceeds almost identically to that of theorem 1.2.53. We first reduce to considering $\mathbb{R}^{d \times (d+1)}$, then find $S^{d(d+1)-1} \subseteq \mathbb{R}^{d \times (d+1)}$, which is homeomorphic to $\mathbb{R}^{d \times (d+1)}/\mathbb{R}_+$. Next, observe that $S^{d(d+1)-1}/O(d)$ is homeomorphic to $\mathbb{S}_{\leq d}^{d+1}/\mathbb{R}_+$ by corollary 1.2.42, which is in turn homeomorphic to the boundary of the ball of dimension $(d+1)(d+2)/2 - 1$, i.e. the sphere of one dimension less than that, by proposition 1.2.49. Then a contradiction once again arises by considering the induced maps on homology. We require $d \geq 4$ here so that $d^2 - 1 > (d+1)(d+2)/2 - 2$ and this last step actually produces a contradiction. For lower values of d , one can use [DLS24]'s proof. $\square$

This result also has a more direct proof.

Proof 2. Assume for the sake of contradiction that we have some canonicalization of $\mathbb{R}^{d \times n}$ and the associated section $s : \mathbb{S}_{\leq d}^n \rightarrow \mathbb{R}^{d \times n}$. $\square$

Proposition 1.2.55. *Let $SO(d)$ act on the left of $\mathbb{R}_*^{d \times d}$ via matrix multiplication and let $q : \mathbb{R}_*^{d \times d} \rightarrow \mathbb{R}_*^{d \times d}/SO(d)$ be the quotient map. Then $\text{secat}^s(q) = 2$.*

Proof. First, observe that by theorem 1.2.53, proposition 1.1.94, remark 1.1.14, and lemma 1.1.19 we have that $\text{secat}^s(q) > 1$. Hence, it remains to show that $\text{secat}^s(q) \leq 2$. By proposition 1.2.40 we know that $\mathbb{R}_*^{d \times d}/O(d)$ is homeomorphic to the (disjoint) union of full rank matrices in $\mathbb{S}_+^d$ and $\mathbb{S}_-^d$ (here denoted S_+ and S_-) via the map $q : \mathbb{R}_*^{d \times d} \rightarrow S_+ \sqcup S_-$ defined as follows:

$$q(A) = \begin{cases} A^T A & \text{if } \det(A) > 0 \\ -A^T A & \text{if } \det(A) < 0 \end{cases}.$$

Observe that this is well-defined since $A^T A$ is always positive semi-definite with $\text{rank}(A^T A) = \text{rank}(A)$, so $A^T A \in S_+$ and $-A^T A \in S_-$. Additionally, q is clearly continuous.

Next, we define $s_+ : S_+ \rightarrow \mathbb{R}^{d \times d}$ by $s_+(A) = A^{\frac{1}{2}}$ and $s_- : S_- \rightarrow \mathbb{R}^{d \times d}$ by $s_-(A) = T(-A)^{\frac{1}{2}}$, where $M^{\frac{1}{2}}$ denotes the matrix square root where T is some orthogonal transformation that flips the sign of its input's determinant, i.e. $T \in O(d) \setminus SO(d)$. Both of these maps are clearly well-defined and continuous by construction. For $A \in S_+$, we then compute

$$q(s_+(A)) = q(A^{\frac{1}{2}}) = (A^{\frac{1}{2}})^T (A^{\frac{1}{2}}) = A$$

and similarly for $A \in S_-$ we have

$$q(s_-(A)) = q(T(-A)^{\frac{1}{2}}) = -(T(-A)^{\frac{1}{2}})^T (T(-A)^{\frac{1}{2}}) = -(-A)^{\frac{1}{2}} T^T T (-A)^{\frac{1}{2}} = A.$$

Therefore both s_+ and s_- are local sections of q on S_+ and S_- respectively, and both S_+ and S_- are open in the quotient by construction so we conclude that $\text{secat}^s(q) \leq 2$. $\square$

We would also like to compute or at least bound the sectional category of the quotients $\mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / \text{SO}(d)$ and $\mathbb{R}^{d \times n} / \text{O}(d)$ in general. To do this, it is useful to consider the Stiefel and Grassmann manifolds as in section 1.1.4.

Remark 1.2.56. (From [BZA24]) The Grassmann manifold $\text{Gr}(n, k)$ can also be realized as a quotient of the Stiefel manifold $\text{St}(n, k)$ by relating matrices that are basis for the same subspace, i.e. those that differ by an orthogonal transformation. This is captured by the action of $\text{O}(k)$ on $\text{St}(n, k)$ by right multiplication, hence $\text{Gr}(n, k) \cong \text{St}(n, k) / \text{O}(k)$.

Both of the the Stiefel and Grassman manifolds are compact, and have corresponding non-compact variants:

Definition 1.2.57. The non-compact Stiefel manifold $\text{St}'(n, k)$ is the set of basis matrices for k -dimensional subspaces of $\mathbb{R}^n$, which can be represented via

$$\text{St}'(n, k) := \{U \in \mathbb{R}^{n \times k} : \text{rank}(U) = k\}$$

Definition 1.2.58. The non-compact Grassmann manifold $\text{Gr}'(n, k)$ is $\text{St}'(n, k) / \text{O}(k)$ where $\text{O}(k)$ acts on the right of $\text{St}'(n, k)$ via matrix multiplication.

Remark 1.2.59. The transposition map $A \mapsto A^T$ gives a homeomorphism between $\mathbb{R}_*^{d \times n}$ and $\text{St}'(n, d)$, hence $\mathbb{R}_*^{d \times n} / \text{O}(d) \cong \text{Gr}'(n, d)$.

In view of remark 1.2.59, we now wish to compute/bound the sectional category of the projection $\pi : \text{St}(n, k) \rightarrow \text{Gr}(n, k)$, treating $\text{St}(n, k)$ as the unit sphere in $\mathbb{R}_*^{d \times n}$. But recall that by corollary 1.1.110 this can be done by computing/bounding the cup length of $\text{Gr}(n, k)$.

Proposition 1.2.60. (From [Pav21] following proposition 3.5) For $n \leq m$, $\text{cup}(\text{Gr}(n, k)) \leq \text{cup}(\text{Gr}(m, k))$.

Proposition 1.2.61. (From [Pav21], proposition 3.3) If $2^s < n \leq 2^{s+1}$ for some integer s , then

$$\text{cup}(\text{Gr}(n, 2)) = \begin{cases} 2^{s+1} - 2 & \text{if } n = 2^s + 1 \\ 2^{s+1} - 1 & \text{otherwise.} \end{cases}$$

Proposition 1.2.62. (From [Sto82], p. 104) If $2^s < n \leq 2^{s+1}$ for some positive integer s , we have

$$\text{cup}(\text{Gr}(n, 3)) = \begin{cases} 2^{s+2} - 3 \cdot 2^{p-1} - 4 & \text{if } n = 2^{s+1} - 2^p + 1 \text{ for some } p \geq 1 \\ 2^{s+2} - 3 \cdot 2^{p-1} - 4 + t & \text{if } n = 2^{s+1} - 2^p + t \\ & \text{for some } p \geq 1, 2 \leq t \leq 2^{p-1} \\ 2^{s+2} - 5 & \text{if } n = 2^{s+1} \end{cases}$$

From this, we have a tight lower bound of $\text{cup}(\text{Gr}(n, 3)) \geq 2n - 5$.

Proposition 1.2.63. Letting s range over the positive integers, we have

$$\text{cup}(\text{Gr}(n, 4)) = \begin{cases} 3 \cdot 2^s - 7 + 2^{r+1} + t & \text{if } n = 2^s + 1 + 2^r + t \\ & \text{for some } 0 \leq r < s, 0 \leq t < 2^r \\ 3 \cdot 2^s - 7 & \text{if } n = 2^s + 1 \end{cases}$$

From this, we have a tight lower bound of $\text{cup}(\text{Gr}(n, 4)) \geq \frac{9}{4}n - 8$.

Proposition 1.2.64. (From [Sto82], p.104) For $s \geq 3$, $\text{cup}(\text{Gr}(2^s + 1, 5)) = 3 \cdot 2^s + 2^{s-2} - 9$.

Corollary 1.2.65. Combining propositions 1.2.64 and 1.2.60, we have that

$$\text{cup}(\text{Gr}(n, 5)) \geq \frac{13}{8}n - \frac{85}{8}$$

For more general k , we have the following bound.

Proposition 1.2.66. (Corollary of [Sto82], p. 103) If $2 \leq k \leq n/2$ and $2^s < n \leq 2^{s+1}$ for some integer s , then

$$\text{cup}(\text{Gr}(n, k)) \geq \begin{cases} 2^{s+1} - 2 & \text{if } n = 2^s + 1 \\ 2^{s+1} - 1 & \text{otherwise.} \end{cases}$$

In turn, this means that $\text{cup}(\text{Gr}(n, k)) \geq n - 1$.

All of these cup length bounds can thus be used to compute lower bounds for $\text{secat}^s(q)$ with the quotient map $q : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / \text{O}(d)$, summarized in the following theorem.

Theorem 1.2.67. Let $n \geq d$ and let $q : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / \text{O}(d)$ be the quotient map. Then

$$\text{secat}(q) \geq \begin{cases} 2n - 5 & \text{if } d = 3 \\ \frac{9}{4}n - 8 & \text{if } d = 4 \\ \frac{13}{8}n - \frac{85}{8} & \text{if } d = 5 \\ n - 1 & \text{if } 6 \leq d \leq \frac{n}{2}. \end{cases}$$

We now turn to the case of $q : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / \text{SO}(d)$. This case turns out to be markedly more difficult to study, as we now must consider the *oriented* Grassmann manifolds, which less is known about.

Definition 1.2.68. The oriented Grassmann manifold $\widetilde{\text{Gr}}(n, k)$ is the space of all oriented k -dimensional vector subspaces of $\mathbb{R}^n$.

Remark 1.2.69. $\widetilde{\text{Gr}}(n, k) \simeq \mathbb{R}_*^{d \times n} / \text{SO}(d)$.

Remark 1.2.70. There is a natural vector bundle over $\widetilde{\text{Gr}}(n, k)$ given by

$$\widetilde{E}(n, k) := \left\{ (P, x) \in \widetilde{\text{Gr}}(n, k) \times \mathbb{R}^n : x \in P \right\}^4$$

with the projection $p : \widetilde{E}(n, k) \rightarrow \widetilde{\text{Gr}}(n, k)$.

Remark 1.2.71. There is a natural projection $\widetilde{\pi} : \text{St}(n, k) \rightarrow \widetilde{\text{Gr}}(n, k)$ where $\widetilde{\pi}(B)$ is the plane $\text{span}(B)$ with orientation given by the order of the basis B .

⁴“ $x \in P$ ” is a slight abuse of notation here, since P is not a set but rather an oriented plane. However, the meaning is still the obvious one: that x lies in the (unoriented) plane specified by P .

Proposition 1.2.72. Let $\tilde{\pi} : \text{St}(n, k) \rightarrow \widetilde{\text{Gr}}(n, k)$ be the natural projection. Then the induced map $\tilde{\pi}^* : H^*(\widetilde{\text{Gr}}(n, k)) \rightarrow H^*(\text{St}(n, k))$ is zero on the Stiefel-Whitney classes $w_i(\widetilde{E}(n, k))$.

Proof. This argument is exactly the same as proposition 1.1.109, except that we no longer have that $H^*(\text{Gr}(n, k); \mathbb{Z}_2)$ is generated by Stiefel-Whitney classes. $\square$

Definition 1.2.73. The height of a cohomology class $x \in H^*(X)$ is

$$ht(x) := \sup \{m \in \mathbb{Z}^+ : x^m \neq 0\}$$

where x^m denotes $\underbrace{x \smile x \smile \cdots \smile x}_{m \text{ times}}$.

Remark 1.2.74. In light of proposition 1.2.72, we have that

$$\text{nil ker } \tilde{\pi}^* \geq \sup_{x \in H^*(\widetilde{\text{Gr}}(n, k))} ht(x).$$

Theorem 1.2.75. Let $n/2 \geq d \geq 3$ and let $q : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / \text{SO}(d)$ be the quotient map. Then

$$\text{secat}(q) \geq \begin{cases} \frac{n-d+6}{2} & \text{if } n-d+3 \geq 7 \text{ is odd and } n-d+3 \neq 9, 11 \\ \frac{n-d+5}{2} & \text{if } n-d+3 \geq 7 \text{ is even and } n-d+3 \neq 10, 12 \\ 5 & \text{if } n-d+3 = 9, 10, 11, 12. \end{cases}$$

Proof. This follows directly from remark 1.2.74 and the proof of Proposition B in [Kor06], where the author computes the heights of some Stiefel-Whitney classes. $\square$

Theorem 1.2.76. Let $q : \mathbb{R}_*^{d \times n} \rightarrow \mathbb{R}_*^{d \times n} / O(d)$ be the quotient map. Then $\text{secat}(q) < \infty$.

Proof. First consider $p : \mathbb{R}_*^{d \times n} / \mathbb{R}_+ \rightarrow (\mathbb{R}_*^{d \times n} / O(d)) / \mathbb{R}_+ \cong \text{Gr}_k(\mathbb{R}^n)$. Recall from before the p is a principle $O(d)$ -bundle, so for all $x \in \text{Gr}_k(\mathbb{R}^n)$ we can find a local section $s_x : U_x \rightarrow \mathbb{R}_*^{d \times n} / \mathbb{R}_+$ of p where $U_x \ni x$ is open. Then by compactness of $\text{Gr}_k(\mathbb{R}^n)$, there exists some finite collection $\{x_i\}_{1 \leq i \leq m} \subseteq \text{Gr}_k(\mathbb{R}^n)$ such that $\text{Gr}_k(\mathbb{R}^n) = \cup_i U_{x_i}$. To conclude, observe that $\mathbb{R}_*^{d \times n} / O(d) = \cup_i \mathbb{R}_+ U_{x_i}$ and we have local sections $r_i : \mathbb{R}_+ U_i \rightarrow \mathbb{R}_*^{d \times n}$ given by $r_i(tx) = ts_i([x])$ where $\|x\| = 1$. Therefore by definition we have that $\text{secat}(q) \leq m < \infty$. $\square$

Theorem 1.2.77. Let $q : \mathbb{R}_*^{d \times n} \rightarrow \mathbb{R}_*^{d \times n} / \text{SO}(d)$ be the quotient map. Then $\text{secat}(q) < \infty$.

Proof. This follows exactly the same way as the previous theorem. $\square$

We might also wish to give an upper bound for the sectional category of the two quotient maps considered above. Unfortunately, I could not find a way to do this.

I think it may actually be impossible to find *any* collection of local sections for the entire space $\mathbb{R}^{d \times n}$. If we have $A \in \mathbb{R}^{d \times n}$ with $\text{rank}(A) \leq d-1$ (maybe $\text{rank}(A) \leq d-2$?) any open set $U \subseteq \mathbb{R}^{d \times n} / O(d)$ containing $O(d)A$ will also contain $O(d)L$ for some $L \in \mathbb{R}_*^{d \times n}$ by density, and this seems like it should cause problems.

1.2.4 Canonicalizing $\mathbb{R}^{d \times n}$ with respect to Σ_n

With an extensive exploration of canonicalizing $\mathbb{R}^{d \times n}$ with respect to rotation, we now follow the lead of [DLS24] and consider canonicalizing $\mathbb{R}^{d \times n}$ with respect to permutations. More precisely, we wish to canonicalize $\mathbb{R}^{d \times n}$ with respect to Σ_n , the symmetric group on n elements. Practically speaking, this would allow us to operate on unordered collections of points in $\mathbb{R}^d$.

Definition 1.2.78. *The symmetric group on n elements, denoted Σ_n , is the set of all bijections on the first n positive integers with the product given by function composition.*

Definition 1.2.79. *For a topological space X and positive integer n ,*

$$\text{Conf}_n(X) := \{(x_1, \dots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}$$

is the space of n ordered mutually distinct points in X .

Definition 1.2.80. *For a topological space X and positive integer n ,*

$$\text{UConf}_n(X) := \text{Conf}_n(X)/\Sigma_n = \{\{x_1, \dots, x_n\} \subseteq X : x_i \neq x_j \text{ for } i \neq j\}$$

is the space of n unordered mutually distinct points in X .

Definition 1.2.81. *A G -action on X is properly discontinuous if for every $x \in X$ there is an open neighborhood $U \ni x$ such that $gU \cap U = \emptyset$ for all $g \in G$.*

Lemma 1.2.82. *Let G be a finite group that acts freely on a Hausdorff space X . Then the action of G is properly discontinuous.*

Proof. Let $x \in X$ and write $G = \{g_1, \dots, g_n\}$ where $g_1 = e_G$. Since X is Hausdorff, we can find open neighborhoods $U_1, \dots, U_n$ of $g_1x, \dots, g_nx$ such that $U_i \cap U_j = \emptyset$ for $i \neq j$.⁵ Now let $U = \bigcap_{i=2}^n g_i^{-1}U_i$ and observe that U is an open set containing x with $g_iU \subseteq U_i$. To finish, let $g \in G$ such that $gU \cap U \neq \emptyset$. Since $g_iU \cap U = \emptyset$ for $i \neq 1$ by construction, we conclude that $g = g_1$ and G acts properly discontinuously on X . $\square$

Lemma 1.2.83. *If X is Hausdorff and $\text{Conf}_n(X)$ is path-connected, then the quotient map $q : \text{Conf}_n(X) \rightarrow \text{UConf}_n(X)$ is a covering space. Additionally, q is an $n!$ -sheeted covering.*

Proof. By proposition 1.40 of [Hat05], it suffices to show that the action of Σ_n on $\text{Conf}_n(X)$ is properly discontinuous, so by lemma 1.2.82 it suffices to show that Σ_n acts freely on $\text{Conf}_n(X)$. However, Σ_n clearly acts freely on $\text{Conf}_n(X)$ by the definition of $\text{Conf}_n(X)$, so q is a covering map.

For the second statement, proposition 1.40 of [Hat05] also gives us that

$$\Sigma_n \cong \pi_1(\text{UConf}_n(X))/q_*(\pi_1(\text{Conf}_n(X))),$$

so by proposition 1.32 of [Hat05] we conclude that the covering has $|\Sigma_n| = n!$ sheets. $\square$

⁵Got construction of U here

Remark 1.2.84. Observe that $\text{Conf}_n(\mathbb{R}^d) \subseteq (\mathbb{R}^d)^n \cong \mathbb{R}^{d \times n}$ and consequently $\text{UConf}_n(\mathbb{R}^d) \subseteq (\mathbb{R}^d)^n / \Sigma_n \cong \mathbb{R}^{d \times n} / \Sigma_n$.

Lemma 1.2.85. For $d \geq 2$, $\text{Conf}_n(\mathbb{R}^d)$ is path-connected.

Proof. Let $x, y \in \text{Conf}_n(\mathbb{R}^d)$ and write $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$. Then for every i , we can find some path

$$\alpha_i : I \rightarrow \mathbb{R}^d \setminus \{y_1, \dots, y_{i-1}, x_{i+1}, \dots, x_n\}$$

such that $\alpha_i(0) = x_i$ and $\alpha_i(1) = y_i$ since $\mathbb{R}^d \setminus A$ for any finite set A is still path-connected. Then we simply construct a path $\gamma : I \rightarrow \text{Conf}_n(\mathbb{R}^d)$ with $\gamma(0) = x$ and $\gamma(1) = y$ by moving coordinate-wise via the α_i 's, i.e.

$$\gamma_i(t) = \begin{cases} x_i & \text{if } 0 \leq t < \frac{i}{n} \\ \alpha_i(nt - i) & \text{if } \frac{i}{n} \leq t < \frac{i+1}{n} \\ y_i & \text{if } \frac{i+1}{n} \leq t \leq 1. \end{cases}$$

We can see that $\gamma(t) \in \text{Conf}_n(\mathbb{R}^d)$ for all $t \in [0, 1]$ since the α_i 's are chosen such that when $\frac{i}{n} \leq t < \frac{i+1}{n}$, we have $\gamma_j(t) = y_j$ for $j < i$, $\gamma_k(t) = x_k$ for $k > i$, and $\gamma_i(t) = \alpha_i(nt - i) \notin \{y_1, \dots, y_{i-1}, x_{i+1}, \dots, x_n\}$. Therefore we conclude that $\text{Conf}_n(\mathbb{R}^d)$ is path-connected. $\square$

Remark 1.2.86. Lemma 1.2.85 fails for $\mathbb{R}$ since removing any points from $\mathbb{R}$ causes it to no longer be path-connected. In greater generality, lemma 1.2.85 holds for any manifold X with $\dim X \geq 2$, and fails if $\dim X = 1$ for the same reasons as for $\mathbb{R}$.

Lemma 1.2.87. Let $p : \tilde{X} \rightarrow X$ be a covering space where $\tilde{X}$ is not homeomorphic to X (i.e. the covering space has more than one sheet). Then there is no global section of p .

Proof. Assume for the sake of contradiction that we have a section $s : X \rightarrow \tilde{X}$ of p , and fix some base point $x \in X$. Since p has more than one sheet, we can find $x_1 \in p^{-1}(x)$ that is distinct from the base point $x_0 \in \tilde{X}$. Then by path-connectivity we can find a path $\gamma : I \rightarrow X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Since $p(x_0) = p(x_1)$ by construction, we have that $p \circ \gamma$ is a loop in X and $[p \circ \gamma]$ is a well-defined element of $\pi_1(X)$. Now observe that p and s induce maps $p_* : \pi_1(\tilde{X}) \rightarrow \pi_1(X)$ and $s_* : \pi_1(X) \rightarrow \pi_1(\tilde{X})$ such that $p_* \circ s_* = \text{id}$. This means we can find $[\alpha] := s_*([p \circ \gamma]) \in \pi_1(\tilde{X})$ such that $p_*([\alpha]) = [p \circ \alpha] = [p \circ \gamma]$, giving us a homotopy $h_t : I \rightarrow X$ such that $h_0 = p \circ \alpha$ and $h_1 = p \circ \gamma$. Since γ clearly lifts $p \circ \gamma$, the homotopy lifting property gives us a unique homotopy lift $\tilde{h}_t : I \rightarrow \tilde{X}$ such that $p \circ \tilde{h}_t = h_t$ with $\tilde{h}_1 = \gamma$. By the uniqueness of lifting we also have that $\alpha = \tilde{h}_0$, but this is impossible since α is a loop and γ is not. Therefore we conclude that there can be no global section s of p . $\square$

We now present a new proof of the uncanonicalizability of $\mathbb{R}^{d \times n}$ with respect to Σ_n . [ZHP20] proved this using a metric argument, but observe that the proof of theorem 1.2.88 generalizes to any manifold of dimension at least 2.

Theorem 1.2.88. If $d, n \geq 2$, there exists no continuous canonicalization of $\mathbb{R}^{d \times n}$ with respect to Σ_n .

Proof. Since $d, n \geq 2$, we immediately conclude by lemmas 1.2.83, 1.2.85, and 1.2.87, as any canonicalization of $\mathbb{R}^{d \times n}$ must descend to a canonicalization of $\text{Conf}_n(\mathbb{R}^d)$. $\square$

Remark 1.2.89. *For completeness, we note that for $d = 1$, sorting is a canonicalization, and for $n = 1$ the identity is a canonicalization.*

As per with rotations, we would like to give some lower bounds for the sectional category of the relevant quotient map here.

Theorem 1.2.90. *(Theorem 1.4 of [Rot08]) Let $q : \text{Conf}_n(\mathbb{R}^d) \rightarrow \text{Conf}_n(\mathbb{R}^d)/\Sigma_n$ be the quotient map, and let $\alpha(n)$ be the number of 1's in the dyadic expansion of n . Then we have*

$$(n - \alpha(n))(d - 1) + 1 \leq \text{secat}(q) \leq \text{cat}(\text{Conf}_n(\mathbb{R}^d)/\Sigma_n) \leq (n - 1)(d - 1) + 1.$$

Corollary 1.2.91. *Let $q : \text{Conf}_n(\mathbb{R}^d) \rightarrow \text{Conf}_n(\mathbb{R}^d)/\Sigma_n$ be the quotient map. Then*

$$\text{secat}(q) \geq (n - \lceil \log_2(n) \rceil)(d - 1) + 1$$

where $\lceil x \rceil$ is the least integer greater than or equal to x .

By theorem 1.2.88 we cannot find a continuous canonicalization of $\mathbb{R}^{d \times n}$ with respect to Σ_n , but theorem 1.2.90 tells us that, if we are willing to restrict to $\text{Conf}_n(\mathbb{R}^d)$ we must be able to find a canonicalization with at most $(n - 1)(d - 1) + 1$ open sets on which the canonicalization is continuous. Unfortunately, the proof of this fact does not appear to be constructive, so the question remains how to produce such a canonicalization that could be used in practice.

1.2.5 Canonicalizing $\mathbb{R}^{d \times n} / \Sigma_n$ with respect to $O(d)$ and $SO(d)$

The last space we might wish to canonicalize is unordered point clouds with respect to rotations. In particular, we wish to produce a section of the quotient $q : \text{UConf}_n(\mathbb{R}^d) \rightarrow \text{UConf}_n(\mathbb{R}^d)/O(d)$, or show that such a map cannot exist. We begin with a few observations.

Lemma 1.2.92. *The actions of Σ_n and $O(d)$ on $\mathbb{R}^{d \times n}$ commute, and we have*

$$(\mathbb{R}^{d \times n} / O(d)) / \Sigma_n \cong (\mathbb{R}^{d \times n} / \Sigma_n) / O(d) \cong \mathbb{R}^{d \times n} / (\Sigma_n \times O(d)).$$

Proof. This follows directly from the associativity of matrix multiplication. Explicitly, let $Q \in O(d)$ and $\sigma \in \Sigma_n$, treating σ as an n by n matrix. Then for any $A \in \mathbb{R}^{d \times n}$,

$$\sigma \cdot Q \cdot A = \sigma \cdot (QA) = (QA)\sigma = Q(A\sigma) = Q \cdot (A\sigma) = Q \cdot \sigma \cdot A.$$

□

Lemma 1.2.93. *The quotient $q : \text{Conf}_n(\mathbb{R}^d) / O(d) \rightarrow (\text{Conf}_n(\mathbb{R}^d) / O(d)) / \Sigma_n$ is an $n!$ -sheeted covering space.*

Proof. Recall by lemma 1.2.82 that it suffices to show that Σ_n acts freely on $\text{Conf}_n(\mathbb{R}^d) / O(d)$. So, let $O(d)A, O(d)B \in \text{Conf}_n(\mathbb{R}^d) / O(d)$ and $\sigma \in \Sigma_n$ such that $O(d)A = (O(d)B)\sigma$. But the actions of $O(d)$ and Σ_n commute so we have $(O(d)B)\sigma = O(d)(B\sigma)$, and we can find $Q \in O(d)$ such that $A = QB\sigma$. Since Σ_n acts freely on $\text{Conf}_n(\mathbb{R}^d)$, we have that $A = QB$. We therefore conclude that $O(d)A = O(d)B$ and q is an $n!$ -sheeted covering space. □

Theorem 1.2.94. *The quotient $q : \text{Conf}_n(\mathbb{R}^d) / O(d) \rightarrow (\text{Conf}_n(\mathbb{R}^d) / O(d)) / \Sigma_n$ has no canonicalization.*

Proof. This follows directly from lemmas 1.2.87 and 1.2.93. □

Corollary 1.2.95. *The quotient $Q : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / (O(d) \times \Sigma_n)$ has no canonicalization.*

Proof. This follows from theorem 1.2.94 and the fact that any section $S : \mathbb{R}^{d \times n} / (O(d) \times \Sigma_n) \rightarrow \mathbb{R}^{d \times n}$ induces a section of $q : \mathbb{R}^{d \times n} / O(d) \rightarrow (\mathbb{R}^{d \times n} / O(d)) / \Sigma_n$ via $s := p \circ S$ where $p : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n} / O(d)$ is the quotient map. □

Note that all the preceding results in this section apply just the same for $SO(d)$ instead of $O(d)$.

Theorem 1.2.96. *Let $X = \{(x_1, \dots, x_n) \in (\mathbb{R}^d)^n : \|x_i\| \neq \|x_j\| \text{ for } i \neq j\}$.*

Chapter 2

Frame Averaging

2.1 Background

2.1.1 Frame Averaging

Definition 2.1.1. Given a G -space X , a frame is a set-valued function $\mathcal{F} : X \rightarrow 2^G \setminus \{\emptyset\}$.

Definition 2.1.2. A frame $\mathcal{F}$ is G -equivariant if for all $g \in G$ and $x \in X$,

$$\mathcal{F}(gx) = g\mathcal{F}(x) = \{gh : h \in \mathcal{F}(x)\}$$

or G -invariant if for all $g \in G$ and $x \in X$,

$$\mathcal{F}(gx) = \mathcal{F}(x).$$

Note that this corresponds to our usual definition of equivariance/invariance with G acting on 2^G element-wise.

Definition 2.1.3. Let X, Y be G -spaces where G acts linearly. A frame $\mathcal{F}$ on X is bounded over $K \subseteq X$ if there exists $c > 0$ such that for all $x \in K$,

$$\sup_{g \in \mathcal{F}(x)} \|g \cdot -\|_{op} = \sup_{g \in \mathcal{F}(x)} \sup_{y \in Y} \|gy\| \leq c.$$

Definition 2.1.4. A frame $\mathcal{F}$ is finite if for all $x \in X$, $|\mathcal{F}(x)|$ is finite. More generally, the cardinality of $\mathcal{F}$ is $|\mathcal{F}| := \sup_{x \in X} |\mathcal{F}(x)|$.

Proposition 2.1.5. (Theorem 1 of [Pun+22]) Let $\mathcal{F}$ be a finite equivariant frame and let $f : X \rightarrow Y$ where Y is a vector space. Then the function $\mathcal{I}_{\mathcal{F}}(f) : X \rightarrow Y$ given by

$$\mathcal{I}_{\mathcal{F}}(f)(x) := \frac{1}{|\mathcal{F}(x)|} \sum_{g \in \mathcal{F}(x)} f(g^{-1}x)$$

is G -invariant.

Proof. Let $x \in X$ and $h \in G$. Then

$$\begin{aligned} \mathcal{I}_{\mathcal{F}}(f)(hx) &= \frac{1}{|\mathcal{F}(hx)|} \sum_{g \in \mathcal{F}(hx)} f(g^{-1}hx) \\ &= \frac{1}{|h\mathcal{F}(x)|} \sum_{g \in h\mathcal{F}(x)} f(g^{-1}hx) \\ &= \frac{1}{|\mathcal{F}(x)|} \sum_{g \in \mathcal{F}(x)} f((hg)^{-1}hx) \\ &= \frac{1}{|\mathcal{F}(x)|} \sum_{g \in \mathcal{F}(x)} f(g^{-1}x) \\ &= \mathcal{I}_{\mathcal{F}}(f)(x). \end{aligned}$$

□

Proposition 2.1.6. (Theorem 2 of [Pun+22]) Let $\mathcal{F}$ be a finite equivariant frame and let $f : X \rightarrow Y$ where Y is a vector space. Then the function $\mathcal{E}_{\mathcal{F}}(f) : X \rightarrow Y$ given by

$$\mathcal{E}_{\mathcal{F}}(f)(x) = \frac{1}{|\mathcal{F}(x)|} g \sum_{g \in \mathcal{F}(x)} f(g^{-1}x)$$

is G -equivariant.

Proof. Let $x \in X$ and $h \in G$. Then

$$\begin{aligned} \mathcal{E}_{\mathcal{F}}(f)(hx) &= \frac{1}{|\mathcal{F}(hx)|} \sum_{g \in \mathcal{F}(hx)} gf(g^{-1}hx) \\ &= \frac{1}{|h\mathcal{F}(x)|} \sum_{g \in h\mathcal{F}(x)} gf(g^{-1}hx) \\ &= \frac{1}{|\mathcal{F}(x)|} \sum_{g \in \mathcal{F}(x)} gf((hg)^{-1}hx) \\ &= \frac{1}{|\mathcal{F}(x)|} \sum_{g \in \mathcal{F}(x)} gf(g^{-1}x) \\ &= \mathcal{E}_{\mathcal{F}}(f)(x). \end{aligned}$$

□

Proposition 2.1.7. (Theorem 3 of [Pun+22]) Let $\mathcal{F}$ be an equivariant frame and let $x \in X$. Then $\mathcal{F}(x)$ is a (disjoint) union of orbits $G_x g \in \mathcal{F}(x)/G_x$.

Proof. Observe that it suffices to show that $hg \in \mathcal{F}(x)$ for all $g \in \mathcal{F}(x)$ and $h \in G_x$. So, fix some $h \in G_x$ and we conclude that $hg \in h\mathcal{F}(x) = \mathcal{F}(hx) = \mathcal{F}(x)$. □

Corollary 2.1.8. Given an equivariant frame $\mathcal{F}$ and function $f : X \rightarrow Y$, we can instead compute the averages in propositions 2.1.5 and 2.1.6 as

$$\mathcal{I}_{\mathcal{F}}(f)(x) = \frac{|G_x|}{|\mathcal{F}(x)|} \sum_{G_x g \in \mathcal{F}(x)/G_x} f(g^{-1}x)$$

and

$$\mathcal{E}_{\mathcal{F}}(f)(x) = \frac{|G_x|}{|\mathcal{F}(x)|} \sum_{G_x g \in \mathcal{F}(x)/G_x} gf(g^{-1}x)$$

respectively.

Corollary 2.1.9. Given an equivariant frame $\mathcal{F}$ and $x \in X$, $|\mathcal{F}(x)| \geq |G_x|$. In particular, the cardinality of $\mathcal{F}$ is at least $\sup_{x \in X} |G_x|$.

Remark 2.1.10. Clearly the largest possible equivariant frame is $\mathcal{F}(x) = G$ for all $x \in X$, so the cardinality of $\mathcal{F}$ is at most $|G|$.

Theorem 2.1.11. (Theorem 2.8 of [DLS24]) Let G be a finite group acting continuously on a locally connected Hausdorff space X , and let $\mathcal{F}$ be an equivariant frame such that $\mathcal{I}_{\mathcal{F}}(f)$ is continuous for every map $f : X \rightarrow Y$ where Y is some non-trivial (real) vector space. If X_{free} is connected, then $\mathcal{F}(x) = G$ for all $x \in \overline{X_{free}}$.

Proof. Define $U = \{x \in X_{free} : e \in \mathcal{F}(x)\}$ and $V = X_{free} \setminus U$. Since G is finite and acts freely on X_{free} , for all $x \in X_{free}$ we can find connected open sets $U_x \ni x$ such that $gU_x \cap U_x = \emptyset$ for $g \neq e$. Fixing some non-zero vector $y \in Y$, there are then continuous maps $f_x : X_{free} \rightarrow Y$ such that $f_x[U_x] = y$ and $f_x[gU_x] = 0$ for $g \neq e$. Fixing $x \in X_{free}$, we know that $\mathcal{I}_{\mathcal{F}}(f_x)$ is continuous, and additionally for $v \in U_x$ we have

$$\begin{aligned} \mathcal{I}_{\mathcal{F}}(f_x)(v) &= \frac{1}{|\mathcal{F}(v)|} \sum_{g \in \mathcal{F}(v)} f_x(g^{-1}v) \\ &= \frac{1}{|\mathcal{F}(v)|} \sum_{g \in \mathcal{F}(v)} \delta_{e,g} y \\ &= \frac{y}{|\mathcal{F}(v)|} \chi_{\mathcal{F}(v)}(e) \end{aligned}$$

Then if $x \in U$ (i.e. $e \in \mathcal{F}(x)$) we have that $\mathcal{I}_{\mathcal{F}}(f_x)(x) = \frac{y}{|\mathcal{F}(x)|}$, and consequently $\mathcal{I}_{\mathcal{F}}(f_x)[U_x] = \frac{y}{|\mathcal{F}(x)|}$ since U_x is connected and $\mathcal{I}_{\mathcal{F}}(f_x)$ is continuous. In particular, this means $e \in \mathcal{F}v$ for all $v \in U_x$, so $U_x \subseteq U$. Then $U = \cup_{x \in U} U_x$, so we have that U is open. Similarly, if $x \in V$ then $\mathcal{I}_{\mathcal{F}}(f_x)[U_x] = 0$ leading to the fact that V is open as well.

Since X_{free} is connected, we must have that either U or V is empty. To see that U is non-empty, let $x \in X_{free}$ be arbitrary and fix some $g \in \mathcal{F}(x)$. Since $\mathcal{F}$ is equivariant we then have that $e \in \mathcal{F}(g^{-1}x)$, so $g^{-1}x \in U$. Hence, we conclude that $U = X_{free}$ and $e \in \mathcal{F}(x)$ for all $x \in X_{free}$.

With this, we can easily show that $\mathcal{F}(x) = G$ for all $x \in X_{free}$. Let $x \in X_{free}$ and $g \in G$, and we know that $e \in \mathcal{F}(g^{-1}x)$ so $g \in \mathcal{F}(x)$ by equivariance of $\mathcal{F}$. Hence, $\mathcal{F}(x) = G$ for all $x \in X_{free}$.

To conclude, let $x \in \overline{X_{free}}$, and we can find a connected open set $U \ni x$ such that $gU \cap U \neq \emptyset$ if and only if $g \in G_x$. Then there is a continuous map $f : X \rightarrow Y$ such that $f[U] = y$ and $f[gU] = 0$ if $g \notin G_x$. Then for $v \in U$ we have as before that

$$\begin{aligned} \mathcal{I}_{\mathcal{F}}(f)(v) &= \frac{1}{|\mathcal{F}(v)|} \sum_{g \in \mathcal{F}(v)} f(g^{-1}v) \\ &= \frac{1}{|\mathcal{F}(v)|} \sum_{g \in \mathcal{F}(v)} \chi_{G_x}(g) y \\ &= \frac{|G_x \cap \mathcal{F}(v)|}{|\mathcal{F}(v)|} y. \end{aligned}$$

Then by continuity and connectivity we have that $\mathcal{I}_{\mathcal{F}}(f)$ is constant on U . Since $x \in \overline{X_{free}}$, let $\{x_d\}_{d \in D} \subseteq X_{free}$ be a net such that $x_d \rightarrow x$. There is then some $d' \in D$ such that for all $d \geq d'$, $x_d \in U$, and $e \in \mathcal{F}(U)$ so $\mathcal{I}_{\mathcal{F}}(f)(x_d) = \frac{|G_x \cap \mathcal{F}(x_d)|}{|\mathcal{F}(x_d)|} y = \frac{|G_x|}{|G|} y \neq 0$. Then by continuity of $\mathcal{I}_{\mathcal{F}}(f)$ we have that $\mathcal{I}_{\mathcal{F}}(f)(x) = \lim_{d \in D} \mathcal{I}_{\mathcal{F}}(f)(x_d) \neq 0$ so $G_x \cap \mathcal{F}(x) \neq \emptyset$.

Letting $g \in G_x \cap \mathcal{F}(x)$, we have that $e \in g^{-1}\mathcal{F}(x) = \mathcal{F}(g^{-1}x) = \mathcal{F}(x)$. This shows that $e \in \mathcal{F}(x)$ for all $x \in \overline{X_{free}}$. Finally, for $x \in \overline{X_{free}}$ and $g \in G$, and we know that $e \in \mathcal{F}(g^{-1}x)$ so $g \in \mathcal{F}(x)$ by equivariance of $\mathcal{F}$. Therefore $\mathcal{F}(x) = G$ for all $x \in \overline{X_{free}}$. $\square$

Corollary 2.1.12. (*Corollary 2.9 in [DLS24]*) *Let $d, n > 1$ and consider Σ_n acting on $\mathbb{R}^{d \times n}$. If $\mathcal{F}$ is a continuity-preserving equivariant frame, then $\mathcal{F}(A) = \Sigma_n$ for all $A \in \mathbb{R}^{d \times n}$. Note that this provides an alternate proof of theorem 1.2.88, since a canonicalization is a frame with cardinality 1.*

Lemma 2.1.13. *Let X be a G -space such that the quotient $q : X \rightarrow X/G$ is a principal G -bundle (for instance, G is a compact Lie group acting freely on X). Then there exists a section $s : X/G \rightarrow X$ of q if and only if there exists a G -equivariant map $\varphi : X \rightarrow_G G$.*

Proof. First, assume we have a section s of q . For two elements $x, y \in X$ in the same G orbit, let $g \in G$ be the unique element such that $gy = x$ (this is well-defined since G acts freely), we then define $x/y = g$. Now define the function $\varphi : X \rightarrow G$ via $\varphi(x) = x/(s \circ q)(x)$. Observe that φ is G -equivariant since $x \mapsto x$ is clearly equivariant and $(s \circ q)(x)$ is invariant. To see that φ is continuous, let $U \subseteq G$ be open. Then

$$\begin{aligned} \varphi^{-1}[U] &= \{x \in X : x/(s \circ q)(x) \in U\} \\ &= \{x \in X : \exists g \in U, g(s \circ q)(x) = x\} \\ &= \{x \in X : (s \circ q)(x) \in U^{-1}x\} \\ &= (s \circ q)^{-1}[U^{-1}x]. \end{aligned}$$

Observe that U^{-1} is open since $(-)^{-1}$ is a homeomorphism on any topological group, then $U^{-1}x$ is open as well since q is a principal G -bundle. Hence, $\varphi^{-1}[U]$ is open since $s \circ q$ is continuous, so we conclude that φ is a continuous equivariant map.

Conversely, assume φ is an equivariant map. Then we can define $s : X/G \rightarrow X$ via $s(Gx) = \varphi(x)^{-1}x$. Observe that s is indeed a well-defined map since

$$s(Ggx) = \varphi(gx)^{-1}gx = g\varphi(x)^{-1}gx = \varphi(x)^{-1}g^{-1}gx = \varphi(x)^{-1}x = s(Gx).$$

Then s is continuous by the continuity of φ and $(-)^{-1}$ and the definition of the quotient topology, so we are done. $\square$

Lemma 2.1.14. *Let X be a G -space with compact G such that $q : X \rightarrow X/G$ is a principal G -bundle and let $F : X \rightarrow G^n/\Sigma_n$. If $\mathcal{I}_F(f)(x) = \frac{1}{n}\sum_{g \in F(x)} f(g^{-1}x)$ is continuous for all $f : X \rightarrow \mathbb{R}$, then F is continuous.*

Proof. Assume for the sake of contradiction there is some $x_0 \in X$ such that F is not continuous, so we can find a net $\{x_d\}_{d \in D} \subseteq X$ with $x_d \rightarrow x_0$ such that $F(x_d) \not\rightarrow F(x_0)$. Since G is compact, without loss of generality we can assume that $F(x_d) \rightarrow L \neq F(x_0)$ by passing to a subnet. Now since q is a principal G -bundle, there exists an open neighborhood U of $q(x_0)$ such that $q^{-1}[U] \cong G \times U$. This homeomorphism is demonstrated by a section $s : X/G \rightarrow X$ of q so we have the map $x \mapsto h \cdot s(q(x))$ for some unique $h \in G$ (this is continuous and well-defined since q is a principal G -bundle). We can then find a continuous function $\varphi : G \rightarrow \mathbb{R}$ such that $\sum_{g \in L} \varphi(g^{-1}h) \neq \sum_{g \in F(x_0)} \varphi(g^{-1}h)$ where $x_0 = h \cdot s(q(x_0))$. Then by Urysohn's

lemma there is some $\psi : X \rightarrow \mathbb{R}$ such that $\psi(x_0) = 1$ and $\psi[X \setminus q^{-1}[U]] = 0$, so the function $f : X \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \psi(x)\varphi(g) & \text{if } x = g \cdot s(q(x)) \in q^{-1}[U] \\ 0 & \text{otherwise} \end{cases}$$

is well-defined and continuous. Then by assumption we have that

$$\mathcal{I}_F(f)(x_d) \rightarrow \mathcal{I}_F(f)(x_0) = \frac{1}{n} \sum_{g \in F(x_0)} f(g^{-1}x_0) = \frac{1}{n} \sum_{g \in F(x_0)} \varphi(g^{-1}h).$$

However, the map from G^n/Σ_n to $\mathbb{R}$ given by

$$\{g_1, \dots, g_n\} \mapsto \frac{1}{n} \sum_i f(g_i^{-1}x)$$

for any fixed $f : X \rightarrow \mathbb{R}$ and $x \in X$ is also continuous by the definition of the quotient topology, so we have

$$\mathcal{I}_F(f)(x_d) = \frac{1}{n} \sum_{g \in F(x_d)} f(g^{-1}x_d) \rightarrow \frac{1}{n} \sum_{g \in L} f(g^{-1}x_0) = \frac{1}{n} \sum_{g \in L} \varphi(g^{-1}h).$$

This is a contradiction since these limits are not equal, so we conclude that F must be continuous. $\square$

Theorem 2.1.15. (Theorem 2.10 of [DLS24]) Let $n \geq 2$ and let $\mathcal{F}$ be a continuity-preserving equivariant frame for $\text{SO}(2)$ acting on $X := \mathbb{R}^{2 \times n} \setminus 0$. Then $|\mathcal{F}| = \infty$.

Definition 2.1.16. Let μ be a Borel measure on a topological group G , and let $g \in G$. Then the pushforward measure $g_*\mu$ is defined for Borel sets $E \subseteq G$ as $g_*\mu(E) := \mu(g^{-1}E)$.

Definition 2.1.17. Let μ be a Borel measure on a topological group G which acts on X , and let $x \in X$. Then the measure $\langle \mu \rangle_x$ is defined for Borel sets $E \subseteq G$ by

$$\langle \mu \rangle_x(E) := \int_{g \in G_x} g_*\mu(E) dg = \int_{g \in G_x} \mu(g^{-1}E) dg$$

where the integral is with respect to the Haar measure of G_x .

Definition 2.1.18. Given a G -space X , a weighted frame is a function $\mu_\bullet : X \rightarrow \mathcal{P}(G)$ where $\mathcal{P}(G)$ is the space of Borel probability measures on G . For $x \in X$ we write $\mu_x = \mu_\bullet(x)$ and $\bar{\mu}_x := \langle \mu_x \rangle_x$.

Definition 2.1.19. A weighted frame $\mu_\bullet$ is equivariant at $x \in X$ if for all $g \in G$, $\mu_{gx} = g_*\mu_x$. $\mu_\bullet$ is equivariant if it is equivariant at all $x \in X$.

Definition 2.1.20. A weighted frame $\mu_\bullet$ is weakly equivariant at $x \in X$ if for all $g \in G$, $\bar{\mu}_{gx} = g_*\bar{\mu}_x$. $\mu_\bullet$ is weakly equivariant if it is weakly equivariant at all $x \in X$.

Remark 2.1.21. *If a weighted frame $\mu_\bullet$ is equivariant at $x \in X$, then it is weakly equivariant at x .*

Proof. Let $g \in G$ and let $E \subseteq G$ be a Borel set. Then

$$\begin{aligned}
\bar{\mu}_{gx}(E) &= \langle \mu_{gx} \rangle_{gx}(E) \\
&= \int_{h \in G_{gx}} h_* \mu_{gx}(E) dh \\
&= \int_{h \in G_{gx}} \mu_{gx}(h^{-1}E) dh \\
&= \int_{h \in G_{gx}} g_* \mu_x(h^{-1}E) dh \\
&= \int_{h \in G_{gx}} \mu_x(g^{-1}h^{-1}E) dh \\
&= \int_{h \in gG_x g^{-1}} \mu_x(g^{-1}h^{-1}E) dh \\
&= \int_{h \in G_x} \mu_x(h^{-1}g^{-1}E) dh \\
&= g_* \int_{h \in G_x} h_* \mu_x(E) dh \\
&= g_* \langle \mu_x \rangle_x(E) \\
&= g_* \bar{\mu}_x(E).
\end{aligned}$$

□

Remark 2.1.22. *If a weighted frame $\mu_\bullet$ is weakly equivariant at $x \in X$ and G_x is trivial, then it is equivariant at x .*

Proof. Let $g \in G$ and let $E \subseteq G$ be a Borel set. Then

$$\begin{aligned}
\mu_{gx}(E) &= e_* \mu_{gx}(E) \\
&= \int_{h \in G_x} h_* \mu_{gx}(E) dh \\
&= \int_{h \in G_{gx}} h_* \mu_{gx}(E) dh \\
&= \bar{\mu}_{gx} \\
&= g_* \bar{\mu}_x \\
&= g_* \int_{h \in G_x} h_* \mu_x(E) dh \\
&= g_* \mu_x(E).
\end{aligned}$$

□

Lemma 2.1.23. *For any Borel function $f : X \rightarrow \mathbb{R}$, we have*

$$\mathcal{I}_\mu(f)(x) = \int_G f(g^{-1}x) d\mu_x(g) = \int_G f(g^{-1}x) d\bar{\mu}_x(g).$$

Proof. We simply compute

$$\begin{aligned} \mathcal{I}_\mu(f)(x) &= \int_G f(g^{-1}x) d\mu_x(g) \\ &= \int_{G_x} \int_G f(g^{-1}x) d\mu_x(g) dh \\ &= \int_{G_x} \int_G f(g^{-1}h^{-1}x) d\mu_x(g) dh \\ &= \int_{G_x} \int_G f(g^{-1}x) dh_* \mu_x(g) dh \\ &= \int_G f(g^{-1}x) d\left(\int_{G_x} h_* \mu_x(g) dh\right) \\ &= \int_G f(g^{-1}x) d\bar{\mu}_x(g). \end{aligned}$$

□

Proposition 2.1.24. *(Proposition 3.3 in [DLS24]) Let $\mu_\bullet$ be a weakly equivariant weight frame. Then for any Borel function $f : X \rightarrow Y$, the function $\mathcal{I}_\mu(f)$ given by*

$$\mathcal{I}_\mu(f)(x) := \int_G f(g^{-1}x) d\mu_x(g)$$

is G -invariant.

Proof. Using lemma 2.1.23, we simply compute

$$\begin{aligned} \mathcal{I}_\mu(f)(hx) &= \int_G f(g^{-1}hx) d\bar{\mu}_{hx}(g) \\ &= \int_G f(g^{-1}hx) dh_* \bar{\mu}_x(g) \\ &= \int_G f(g^{-1}hx) d\bar{\mu}_x(h^{-1}g) \\ &= \int_G f(hg^{-1}hx) d\bar{\mu}_x(g) \\ &= \int_G f(g^{-1}x) d\bar{\mu}_x(g) \\ &= \mathcal{I}_\mu(f)(x). \end{aligned}$$

□

Definition 2.1.25. A weakly equivariant weighted frame $\mu_\bullet$ continuous at x if given any sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ converging to x , we have $\langle \mu_{x_n} \rangle_x \rightarrow \langle \mu_x \rangle_x$, i.e. for all $f : G \rightarrow \mathbb{R}$

$$\int_G f d\langle \mu_{x_n} \rangle_x \rightarrow \int_G f d\langle \mu_x \rangle_x = \int_G f d\bar{\mu}_x$$

or equivalently

$$\int_{G_x} \int_G f dh_* \mu_{x_n} dh \rightarrow \int_{G_x} \int_G f dh_* \mu_x dh.$$

A robust frame is a continuous weakly equivariant weighted frame.

Lemma 2.1.26. ¹ Let K and X be topological spaces with K compact, and let $f : K \times X \rightarrow \mathbb{R}$ be continuous. Then fixing some $x_0 \in X$, the map $M : X \rightarrow \mathbb{R}$ given by

$$M(x) := \max_{k \in K} |f(k, x_0) - f(k, x)|$$

is continuous.

Proof. First, define the map $D : K \times X \rightarrow \mathbb{R}$ by $D(k, x) := |f(k, x_0) - f(k, x)|$, which is clearly continuous, and note that $M(x) = \max_{k \in K} D(k, x)$. To show that M is continuous we will show that it is continuous at all $x \in X$. So, fix some $x \in X$ and let $\varepsilon > 0$. By continuity of D , for each $k \in K$ we can find an open set of the form $U_k \times V_k \subseteq K \times X$ such that $(k, x) \in U_k \times V_k$ and $|D(k, x) - D(h, y)| < \varepsilon/2$ for all $(h, y) \in U_k \times V_k$.

Clearly $K = \cup_{k \in K} U_k$, so by compactness we can find finitely many $k_1, \dots, k_n \in K$ such that $K = \cup_{i \leq n} U_{k_i}$. We then define $V = \cap_{i \leq n} V_{k_i}$, which is open and non-empty since $x \in V_{k_i}$ for all $i \leq n$. Now fix $y \in V$ and let $h \in K$, and observe that there is some k_i such that $h \in U_{k_i}$ so we have that $(h, y) \in U_{k_i} \times V \subseteq U_{k_i} \times V_{k_i}$ and hence

$$|D(h, x) - D(h, y)| \leq |D(h, x) - D(k_i, x)| + |D(k_i, x) - D(h, y)| < \varepsilon.$$

This means that for all $y \in V$ we have $\max_{k \in K} |D(k, x) - D(k, y)| < \varepsilon$ and hence

$$|M(x) - M(y)| = \left| \max_{k \in K} D(k, x) - \max_{k \in K} D(k, y) \right| \leq \max_{k \in K} |D(k, x) - D(k, y)| < \varepsilon.$$

We therefore conclude that M is continuous at x , and since x was arbitrary it follows that M is continuous on all of X . $\square$

Lemma 2.1.27. Let $\mu_\bullet$ be a frame such that $\mathcal{I}_\mu$ preserves continuity. Then for any continuous function $f : X \rightarrow \mathbb{R}$, we have

$$\int_G f(g^{-1}x) d\mu_{x_n} \rightarrow \int_G f(g^{-1}x) d\mu_x.$$

¹I couldn't find a standard reference for this lemma, but it is fairly elementary and a special case of Berge's Maximum Theorem. Additionally, Lemma C.1 of [DK26] proves the case where X is a metric space.

Proof. We first compute

$$\begin{aligned}
& \left| \int_G f(g^{-1}x) d\mu_{x_n} - \int_G f(g^{-1}x) d\mu_x \right| \\
& \leq \left| \int_G f(g^{-1}x) d\mu_{x_n} - \int_G f(g^{-1}x_n) d\mu_{x_n} \right| + \left| \int_G f(g^{-1}x_n) d\mu_{x_n} - \int_G f(g^{-1}x) d\mu_x \right| \\
& = \left| \int_G (f(g^{-1}x) - f(g^{-1}x_n)) d\mu_{x_n} \right| + |\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| \\
& \leq \int_G |f(g^{-1}x) - f(g^{-1}x_n)| d\mu_{x_n} + |\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| \\
& \leq \mu_{x_n}(G) \sup_{g \in G} |f(g^{-1}x) - f(g^{-1}x_n)| + |\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| \\
& = \max_{g \in G} |f(g^{-1}x) - f(g^{-1}x_n)| + |\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)|.
\end{aligned}$$

Now clearly $|\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| \rightarrow 0$ since $\mathcal{I}_\mu$ preserves continuity, and by lemma 2.1.26 we also know that

$$\max_{g \in G} |f(g^{-1}x) - f(g^{-1}x_n)| \rightarrow \max_{g \in G} |f(g^{-1}x) - f(g^{-1}x)| = 0$$

since $(g, x) \mapsto f(g^{-1}x)$ is continuous and G is compact. We therefore conclude as desired that

$$\left| \int_G f(g^{-1}x) d\mu_{x_n} - \int_G f(g^{-1}x) d\mu_x \right| \rightarrow 0.$$

□

Theorem 2.1.28. (*Proposition 3.6 in [DLS24]*) *Let G be a compact group acting continuously on a normal Hausdorff space X . A frame $\mu_\bullet$ is continuous if and only if $\mathcal{I}_\mu$ preserves continuity.*

Proof. For the forward direction, assume $\mu_\bullet$ is continuous and let $x \in X$ with $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ such that $x_n \rightarrow x$; we wish to show that $\mathcal{I}_\mu(f)(x_n) \rightarrow \mathcal{I}_\mu(f)(x)$ for any function $f : X \rightarrow \mathbb{R}$.

$$\begin{aligned}
|\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| &= \left| \int_G f(g^{-1}x_n) d\mu_{x_n}(g) - \int_G f(g^{-1}x) d\mu_x(g) \right| \\
&= \left| \int_G f(g^{-1}x_n) d\mu_{x_n}(g) - \int_G f(g^{-1}x) d\langle \mu_x \rangle_x(g) \right| \\
&\leq \left| \int_G f(g^{-1}x_n) d\mu_{x_n}(g) - \int_G f(g^{-1}x) d\langle \mu_{x_n} \rangle_x(g) \right| \\
&\quad + \left| \int_G f(g^{-1}x) d\langle \mu_{x_n} \rangle_x(g) - \int_G f(g^{-1}x) d\langle \mu_x \rangle_x(g) \right|.
\end{aligned}$$

Now observe that the latter term in this last expression has

$$\left| \int_G f(g^{-1}x) d\langle \mu_{x_n} \rangle_x(g) - \int_G f(g^{-1}x) d\langle \mu_x \rangle_x(g) \right| \rightarrow 0$$

since $\mu_\bullet$ is robust, so it remains to study the former term. Then we have

$$\begin{aligned}
\int_G f(g^{-1}x) d\langle \mu_{x_n} \rangle_x(g) &= \int_{G_x} \int_G f(g^{-1}x) dh_* \mu_{x_n}(g) dh \\
&= \int_{G_x} \int_G f(g^{-1}h^{-1}x) d\mu_{x_n}(g) dh \\
&= \int_{G_x} \int_G f(g^{-1}x) d\mu_{x_n}(g) dh \\
&= \int_G f(g^{-1}x) d\mu_{x_n}(g)
\end{aligned}$$

so the former term is

$$\begin{aligned}
\left| \int_G f(g^{-1}x_n) d\mu_{x_n}(g) - \int_G f(g^{-1}x) d\langle \mu_{x_n} \rangle_x(g) \right| &= \left| \int_G f(g^{-1}x_n) d\mu_{x_n}(g) - \int_G f(g^{-1}x) d\mu_{x_n}(g) \right| \\
&= \left| \int_G (f(g^{-1}x_n) - f(g^{-1}x)) d\mu_{x_n}(g) \right| \\
&\leq \int_G |f(g^{-1}x_n) - f(g^{-1}x)| d\mu_{x_n}(g) \\
&\leq \mu_{x_n}(G) \max_{g \in G} |f(g^{-1}x_n) - f(g^{-1}x)| \\
&= \max_{g \in G} |f(g^{-1}x_n) - f(g^{-1}x)| \\
&\rightarrow 0.
\end{aligned}$$

Therefore we conclude that $|\mathcal{I}_\mu(f)(x_n) - \mathcal{I}_\mu(f)(x)| \rightarrow 0$ and $\mathcal{I}_\mu(f)$ is continuous.

Conversely, assume $\mathcal{I}_\mu$ preserves continuity and let $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ be some sequence with $x_n \rightarrow x \in X$. We now wish to show that

$$\int_G f d\langle \mu_{x_n} \rangle_x \rightarrow \int_G f d\langle \mu_x \rangle_x$$

for any continuous $f : G \rightarrow \mathbb{R}$. To see this, define $f_x : G \rightarrow \mathbb{R}$ by

$$f_x(g) := \int_{G_x} f(sg) ds$$

where the integral is with respect to the right-invariant Haar measure on G_x . Observe that f_x induces a map $\widehat{f}_x : G/G_x \rightarrow \mathbb{R}$ since for any $h \in G_x$ we have

$$f_x(hg) = \int_{G_x} f(shg) ds = \int_{G_x} f(sg) ds = f_x(g)$$

where the second equality is by the right-invariance of the Haar measure. We now compute

$$\begin{aligned}
\int_G f d\langle \mu_{x_n} \rangle &= \int_{G_x} \int_G f ds_* \mu_{x_n} ds \\
&= \int_{G_x} \int_G f(sg) d\mu_{x_n} ds \\
&= \int_G \int_{G_x} f(sg) ds d\mu_{x_n} \\
&= \int_G f_x d\mu_{x_n} \\
&= \int_G (\widehat{f}_x \circ q_x) d\mu_{x_n}
\end{aligned}$$

where $q_x : G \rightarrow G/G_x$ is the quotient map.

Next, we define a map $F : Gx \rightarrow \mathbb{R}$ by $F(gx) := f_x(g^{-1})$. Note that F is indeed well-defined since for any $h \in G_x$, we have

$$F(ghx) = f_x(h^{-1}g^{-1}) = \int_{G_x} f(sh^{-1}g^{-1}) ds = \int_{G_x} f(sg^{-1}) ds = F(gx)$$

where the third equality is from the right-invariance of the Haar measure on G_x . Since Gx is closed (Gx is compact and X is Hausdorff), the Tietze Extension Theorem guarantees the existence of some map $F' : X \rightarrow \mathbb{R}$ such that $F'|_{Gx} = F$; for simplicity, we will not relabel and just consider F as a map from X to $\mathbb{R}$. Observe that by construction we have

$$\int_G F(g^{-1}x) d\mu = \int_G f_x(g) d\mu$$

for any measure μ on G , so by lemma 2.1.27 we conclude that

$$\int_G f d\langle \mu_{x_n} \rangle_x = \int_G f_x d\mu_{x_n} = \int_G F(g^{-1}x) d\mu_{x_n} \rightarrow \int_G F(g^{-1}x) d\mu_x = \int_G f_x d\mu_x = \int_G f d\langle \mu_x \rangle_x.$$

Therefore $\mu_\bullet$ is continuous. □

2.1.2 Canonicalization Frames

This material comes primarily from [Ma+24].

Definition 2.1.29. Given a G -set X , a canonicalization frame is a G -invariant function $\mathcal{C} : X \rightarrow 2^X \setminus \{\emptyset\}$.

Definition 2.1.30. A canonicalization frame $\mathcal{C}$ is invariant if for all $x \in X$ and $g \in G$, $\mathcal{C}(gx) = \mathcal{C}(x)$.

Definition 2.1.31. A canonicalization frame $\mathcal{C}$ is orbital if for all $x \in X$, $\mathcal{C}(x) \subseteq Gx$.

Proposition 2.1.32. Given a canonicalization frame $\mathcal{C}$ and map $f : X \rightarrow \mathbb{R}$, the function $\mathcal{I}_{\mathcal{C}}(f)$ given by

$$\mathcal{I}_{\mathcal{C}}(f)(x) = \frac{1}{|\mathcal{C}(x)|} \sum_{c \in \mathcal{C}(x)} f(c)$$

is G -invariant.

Proof. Let $x \in X$ and $g \in G$, and we immediately have

$$\mathcal{I}_{\mathcal{C}}(f)(gx) = \frac{1}{|\mathcal{C}(gx)|} \sum_{c \in \mathcal{C}(gx)} f(c) = \frac{1}{|\mathcal{C}(x)|} \sum_{c \in \mathcal{C}(x)} f(c) = \mathcal{I}_{\mathcal{C}}(f)(x).$$

□

Theorem 2.1.33. (Theorem 3.1 of [Ma+24]) For every equivariant frame $\mathcal{F}$, there exists a canonicalization frame $\mathcal{C}$ such that $\mathcal{I}_{\mathcal{F}} = \mathcal{I}_{\mathcal{C}}$. Conversely, for every canonicalization frame $\mathcal{C}$ there exists an equivariant frame $\mathcal{F}$ such that $\mathcal{I}_{\mathcal{C}} = \mathcal{I}_{\mathcal{F}}$. In both cases, for all $x \in X$ we have

$$|\mathcal{C}(x)| |G_x| = |\mathcal{F}(x)|.$$

Proof. To begin, let $\mathcal{F}$ be an equivariant frame. Then for all $x \in X$, we define $\mathcal{C}(x) = \{gx : g \in \mathcal{F}(x)\}$, which is invariant since

$$\mathcal{C}(hx) = \{ghx : g \in \mathcal{F}(hx)\} = \{ghx : g \in h\mathcal{F}(x)\} = \{gx : g \in \mathcal{F}(x)\} = \mathcal{C}(x).$$

For the other direction, let $\mathcal{C}$ be a canonicalization frame. Then for all $x \in X$, we define $\mathcal{F}(x) = \{g \in G : g^{-1}x \in \mathcal{C}(x)\}$, which is equivariant since

$$\begin{aligned} \mathcal{F}(hx) &= \{g \in G : g^{-1}hx \in \mathcal{C}(hx)\} \\ &= \{g \in G : g^{-1}hx \in \mathcal{C}(x)\} \\ &= h \{g \in G : g^{-1}x \in \mathcal{C}(x)\} \\ &= h\mathcal{F}(x). \end{aligned}$$

In both cases, it follows from proposition 2.1.7 that $\mathcal{I}_{\mathcal{C}} = \mathcal{I}_{\mathcal{F}}$ and $|\mathcal{C}(x)| |G_x| = |\mathcal{F}(x)|$. □

Corollary 2.1.34. Let G be a finite group acting continuously on a locally connected Hausdorff space X , and let $\mathcal{C}$ be an equivariant frame such that $\mathcal{I}_{\mathcal{C}}(f)$ is continuous for every map $f : X \rightarrow Y$ where Y is some non-trivial (real) vector space. If X_{free} is connected, then $\mathcal{C}(x) = G$ for all $x \in \overline{X_{free}}$.

Proof. This follows directly from theorems 2.1.11 and 2.1.33 since $|G_x| = 1$ for all $x \in X$. $\square$

Corollary 2.1.35. *Let $n \geq 2$ and let $\mathcal{C}$ be a continuity-preserving canonicalization frame for $\mathrm{SO}(2)$ acting on $X := \mathbb{R}^{2 \times n} \setminus 0$. Then $|\mathcal{C}| = \infty$.*

Proof. This follows directly from theorems 2.1.15 and 2.1.33 since $|G_x| = 1$ for all $x \in X$. $\square$

2.2 New Work

2.2.1 Extending Canonicalization Frames

Following [DLS24], we make the following definitions in the context of canonicalization frames.

Definition 2.2.1. *Given a G -space X , a weighted canonicalization frame is a function $\nu_\bullet : X \rightarrow \mathcal{P}(X)$ where $\mathcal{P}(X)$ is the space of Borel probability measures on X .*

Definition 2.2.2. *A weighted canonicalization frame $\nu_\bullet$ is orbital if $\nu_x(X \setminus Gx) = 0$ for all $x \in X$.*

Remark 2.2.3. *As per [DLS24], definition 2.2.2 makes sense for arbitrary measures, but we will be primarily concerned with finite atomic measures of the form $\nu_x = \sum_i \alpha_i \delta_{x_i}$ where $\alpha_i \in [0, 1]$ and $\sum_i \alpha_i = 1$.*

Definition 2.2.4. *A weighted canonicalization frame $\nu_\bullet$ is invariant if $\nu_{gx} = \nu_x$ for all $x \in X$ and $g \in G$.*

Proposition 2.2.5. *Let $\nu_\bullet$ be an invariant weighted canonicalization frame. Then for any Borel function $f : X \rightarrow \mathbb{R}$, the function $\mathcal{I}_\nu(f)$ given by*

$$\mathcal{I}_\nu(f)(x) := \int_X f d\nu_x$$

is G -invariant.

Proof. Let $g \in G$ and $x \in X$, and we simply compute

$$\mathcal{I}_\nu(f)(gx) = \int_X f d\nu_{gx} = \int_X f d\nu_x = \mathcal{I}_\nu(f)(x).$$

□

Definition 2.2.6. *An invariant orbital weighted canonicalization frame $\nu_\bullet$ is continuous at x if for any sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ converging to x , we have $\nu_{x_n} \rightarrow \nu_x$, i.e. for all $f : X \rightarrow \mathbb{R}$*

$$\int_X f d\nu_{x_n} \rightarrow \int_X f d\nu_x.$$

A robust canonicalization frame is a continuous invariant orbital weighted canonicalization frame.

Theorem 2.2.7. *A weighted canonicalization frame $\nu_\bullet$ is continuous if and only if $\mathcal{I}_\nu$ preserves continuity.*

Proof. First assume that $\nu_\bullet$ is continuous and let $x \in X$ with $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ such that $x_n \rightarrow x$. Then for any map $f : X \rightarrow \mathbb{R}$, we immediately have

$$\mathcal{I}_\nu(f)(x_n) = \int_X f d\nu_{x_n} \rightarrow \int_X f d\nu_x = \mathcal{I}_\nu(f)(x)$$

so we conclude that $\mathcal{I}_\nu$ preserves continuity.

Now instead assume that $\mathcal{I}_\nu$ preserves continuity, let $f : X \rightarrow \mathbb{R}$ be a Borel function, and let $x \in X$ with $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ such that $x_n \rightarrow x$. Then by continuity of $\mathcal{I}_\nu(f)$ we have

$$\int_X f d\nu_{x_n} = \mathcal{I}_\nu(f)(x_n) \rightarrow \mathcal{I}_\nu(f)(x) = \int_X f d\nu_x$$

so we are done. $\square$

Similar to theorem 2.1.33, we have a correspondence between weighted frames and weighted canonicalization frames.

Theorem 2.2.8. *Let X be a normal G -space. Then every weighted frame induces a weighted canonicalization frame and vice versa (though not necessarily via a bijection). In particular, this correspondence sends weakly equivariant weighted frames to invariant orbital weighted canonicalization frames and weighted canonicalization frames to equivariant weighted frames, and both directions preserve the continuity of the frame.*

Proof. To begin, let $\mu_\bullet$ be a weakly equivariant weighted frame. We define a weighted canonicalization frame $\nu_\bullet$ for Borel sets $E \subseteq X$ by

$$\nu_x(E) = \bar{\mu}_x(\{g \in G : g^{-1}x \in E\}).$$

Then $\nu_\bullet$ is orbital since

$$\nu_x(X \setminus Gx) = \bar{\mu}_x(\{g \in G : g^{-1}x \notin Gx\}) = \bar{\mu}_x(\emptyset) = 0$$

and invariant since

$$\begin{aligned} \nu_{hx}(E) &= \bar{\mu}_{hx}(\{g \in G : g^{-1}hx \in E\}) \\ &= \bar{\mu}_x(h^{-1}\{g \in G : g^{-1}hx \in E\}) \\ &= \bar{\mu}_x(\{g \in G : (hg)^{-1}hx = g^{-1}x \in E\}) \\ &= \nu_x(E). \end{aligned}$$

Now assume $\mu_\bullet$ is continuous, and we wish to show that $\mathcal{I}_\nu(f)$ is continuous for any continuous f . To see this, first let $E \subseteq X$ be a Borel set and observe

$$\int_X \chi_E d\nu_x = \nu_x(E) = \bar{\mu}_x(\{g \in G : g^{-1}x \in E\}) = \int_G \chi_E(g^{-1}x) d\bar{\mu}_x(g).$$

Since any Borel function $f : X \rightarrow \mathbb{R}$ can be approximated by simple functions, the Lebesgue Dominated Convergence Theorem gives us that

$$\int_X f d\nu_x = \int_G f(g^{-1}x) d\bar{\mu}_x(g).$$

But recall by lemma 2.1.23 that

$$\int_G f(g^{-1}x) d\bar{\mu}_x(g) = \int_G f(g^{-1}x) d\mu_x(g)$$

so we have that $\mathcal{I}_\nu(f) = \mathcal{I}_\mu(f)$ and hence $\nu_\bullet$ is continuous by theorem 2.2.7.

For the other direction, let $\nu_\bullet$ be an invariant orbital weighted canonicalization frame. We define a weighted frame $\mu_\bullet$ as follows.² To begin, fix some $x \in X$ and define a linear function $T_x : C(G; \mathbb{R}) \rightarrow C(G/G_x; \mathbb{R})$ via

$$T_x(f)(gG_x) := \int_{G_x} f((gs)^{-1})ds$$

where this integral is with respect to the unique left-invariant normalized Haar measure on G_x . To verify that this is a well-defined function, let $g' \in G$ such that $g'G_x = gG_x$, i.e. $g' = gs'$ for some $s' \in G_x$. Then we have

$$T_x(f)(g'G_x) = \int_{G_x} f((g's)^{-1})ds = \int_{G_x} f((gs's)^{-1})ds = \int_{G_x} f((gs)^{-1})ds = T_x(f)(gG_x)$$

where the third equality is by the left-invariance of the Haar measure. Observe that T_x is also linear by the linearity of integration. Using T_x , we now define a linear functional $L_x : C(G/G_x; \mathbb{R}) \rightarrow \mathbb{R}$ via

$$L_x(f) := \int_{G_x} f \circ q_x d\nu_x$$

where $q_x : Gx \rightarrow G/G_x$ is the quotient map. Observe that L_x is clearly linear, again by the linearity of integration. Then $L_x \circ T_x : C(G; \mathbb{R}) \rightarrow \mathbb{R}$ is a linear functional since L_x and T_x are both linear, and G is compact, so the Riesz-Markov-Kakutani Representation Theorem gives us that there is a unique Borel measure μ_x on G such that for any Borel function $f : G \rightarrow \mathbb{R}$,

$$\int_G f d\mu_x = (L_x \circ T_x)(f).$$

Define $\mu_\bullet$ as the collection of these μ_x 's, and note that μ_x is indeed a probability measure since

$$T_x(1)(gG_x) = \int_{G_x} 1ds = 1$$

so

$$\mu_x(G) = \int_G 1d\mu_x = (L_x \circ T_x)(1) = \int_{Gx} (T_x(1) \circ q_x) d\nu_x = \int_{Gx} 1d\nu_x = 1$$

²The idea for this construction was suggested in part by Claude via this chat.

To see that $\mu_\bullet$ is equivariant, let $h \in G$ and $x \in X$. Then we compute

$$\begin{aligned}
\int_G f d\mu_{hx} &= \int_{G_{hx}} T_{hx}(f)(q_{hx}(c)) d\nu_{hx}(c) \\
&= \int_{G_x} T_{hx}(f)(q_{hx}(c)) d\nu_x(c) \\
&= \int_{G_x} \int_{G_{hx}} f((gs)^{-1}) ds d\nu_x(c) & (\exists g \in G, c = ghx \implies q_{hx}(c) = gG_{hx}) \\
&= \int_{G_x} \int_{hG_x h^{-1}} f((gs)^{-1}) ds d\nu_x(c) \\
&= \int_{G_x} \int_{G_x} f((ghsh^{-1})^{-1}) ds d\nu_x(c) \\
&= \int_{G_x} \int_{G_x} f(h((gh)s)^{-1}) ds d\nu_x(c) \\
&= \int_{G_x} (T_x(f \circ (h \cdot -))(q_x(c))) d\nu_x(c) & (c = ghx \implies q_x(c) = ghG_x) \\
&= \int_G (f \circ (h \cdot -)) d\mu_x \\
&= \int_G f dh_* \mu_x
\end{aligned}$$

and we thus conclude that $\mu_{hx} = h_* \mu_x$. Lastly, assume $\nu_\bullet$ is continuous, and we wish to show that $\mathcal{I}_\mu(f)$ is continuous for any map $f : X \rightarrow \mathbb{R}$. To see this, we simply compute

$$\begin{aligned}
\mathcal{I}_\mu(f)(x) &= \int_G f(g^{-1}x) d\mu_x \\
&= \int_G f_x^{-1}(g) d\mu_x & (f_x^{-1}(g) := f(g^{-1}x)) \\
&= (L_x \circ T_x)(f_x^{-1}) \\
&= \int_{G_x} (T_x(f_x^{-1}) \circ q_x)(c) d\nu_x(c) \\
&= \int_{G_x} \int_{G_x} f_x^{-1}((gs)^{-1}) ds d\nu_x(c) & (c = gx) \\
&= \int_{G_x} \int_{G_x} f(gsx) ds d\nu_x(c) \\
&= \int_{G_x} \int_{G_x} f(gx) ds d\nu_x(c) \\
&= \int_{G_x} f(c) d\nu_x(c) \\
&= \int_X f(c) d\nu_x(c) \\
&= \mathcal{I}_\nu(f).
\end{aligned}$$

This means that $\mathcal{I}_\mu$ preserves continuity, so we have that $\mu_\bullet$ is continuous by theorem 2.1.28.

Lastly, we will show that each direction of this process is (essentially) invertible. First, consider some weakly equivariant weighted frame $\mu_\bullet$; let $\nu_\bullet$ be the invariant orbital weight canonicalization frame induced by $\mu_\bullet$; and let $\lambda_\bullet$ be the equivariant weighted frame induced by $\nu_\bullet$. We now wish to show that $\lambda_x = \bar{\mu}_x$ for any $x \in X$, so let $f : G \rightarrow \mathbb{R}$ be Borel and observe

$$\begin{aligned}
\int_G f d\lambda_x &= \int_{G_x} (T_x(f) \circ q_x) d\nu_x \\
&= \int_G (T_x(f) \circ q_x)(g^{-1}x) d\bar{\mu}_x \\
&= \int_G (T_x(f) \circ q_x)(g^{-1}x) d\mu_x && \text{(lemma 2.1.23)} \\
&= \int_G (T_x(f) \circ q_x)(g^{-1}x) d\mu_x \\
&= \int_G \int_{G_x} f((g^{-1}s)^{-1}) ds d\mu_x \\
&= \int_G \int_{G_x} f(s^{-1}g) ds d\mu_x \\
&= \int_G \int_{G_x} f(sg) ds d\mu_x \\
&= \int_G f d\bar{\mu}_x.
\end{aligned}$$

Since f was arbitrary, we have that $\lambda_x = \bar{\mu}_x$.

For the other direction, consider some invariant orbital weighted canonicalization frame $\nu_\bullet$; let $\mu_\bullet$ be the equivariant weight frame induced by $\nu_\bullet$; and let $v_\bullet$ be the invariant orbital weighted canonicalization frame induced by $\mu_\bullet$. We now wish to show that $v_x = \nu_x$ for any $x \in X$, so let $f : X \rightarrow \mathbb{R}$ be Borel and observe

$$\begin{aligned}
\int_X f dv_x &= \int_G f(g^{-1}x) d\bar{\mu}_x \\
&= \int_G f(g^{-1}x) d\mu_x \\
&= \mathcal{I}_\mu(f)(x) \\
&= \mathcal{I}_\nu(f)(x) \\
&= \int_X f d\nu_x.
\end{aligned}$$

Since f was arbitrary, we have that $v_x = \nu_x$. □

Remark 2.2.9. *If the action of G is free, the correspondence in theorem 2.1.33 is much simpler since every integral over G_x reduces to a single value.*

Theorem 2.2.10 (Richter-Tchakaloff). *(Theorem 1.24 of [Sch17]) Let (X, μ) be a measure space and let $V \subseteq L^1(X)$ be a finite dimensional subspace with $\dim V = k$. Then there exists a k -atomic measure $\nu = \sum_{i \leq k} \alpha_i \delta_{x_i}$ on X such that for every $f \in V$,*

$$\int_X f d\mu = \int_X f d\nu = \sum_{i \leq k} \alpha_i f(x_i).$$

Corollary 2.2.11. *Let X be a G -space where X and G are equipped with some Borel measures ν and μ respectively, and let $V \subseteq L^1(X)$ be a finite dimensional subspace with $\dim V = k$. Then for each $x \in X$, there exists a k -atomic measure $\nu_x = \sum_{i \leq k} \alpha_i \delta_{x_i}$ on G such that for every $f \in V$,*

$$\int_G f(g^{-1}x) d\mu = \int_G f(g^{-1}x) d\nu_x = \sum_{i \leq k} \alpha_i f(g^{-1}x_i).$$

Proof. Since V is finite dimensional, we can find $\{e_i\}_{i \leq k} \subseteq L^1(X)$ with $\text{span}\{e_i\}_{i \leq k} = V$. We can then define $\widehat{e}_i \in L^1(G)$ for $i \leq k$ by $\widehat{e}_i(g) := e_i(g^{-1}x)$. Now let $f \in V$, writing $f = \sum_{i \leq k} \alpha_i e_i$, and define $\widehat{f}(g) = f(g^{-1}x)$. So then setting $W := \text{span}\{\widehat{e}_i\} \subseteq L^1(G)$, we have that $\dim W \leq \dim V$. We then have

$$\widehat{f}(g) = f(g^{-1}x) = \sum_{i \leq k} \alpha_i e_i(g^{-1}x) = \sum_{i \leq k} \alpha_i \widehat{e}_i(g),$$

so $\widehat{f} \in W$ for any $f \in V$. Hence, by theorem 2.2.10 there exists some k -atomic measure $\nu_x = \sum_i \alpha_i \delta_{g_i}$ on G such that

$$\int_G f(g^{-1}x) d\mu = \int_G \widehat{f}(g) d\mu = \int_G \widehat{f}(g) d\nu_x = \sum_i \alpha_i \widehat{f}(g_i) = \sum_i \alpha_i f(g_i^{-1}x).$$

□

Remark 2.2.12. *Note that the measures ν_x produced in corollary 2.2.11 can be taken to be a weighted frame $v_\bullet$ which is automatically continuous. Additionally, this $v_\bullet$ can be made equivariant by simply fixing an element x of each orbit and defining $\nu_{hx} = h_* \nu_x$ for each $h \in G$.*

2.2.2 Weighted Frames and Equivariant Maps into Simplices

In this section, we will explore the duality between (continuous) weighted frames and certain equivariant maps into simplices. In particular, consider the setting of a G -space X where G is finite. Then $\mathcal{P}(G)$, the space of probability measures of G , is just the $(|G| - 1)$ -simplex, since we can label the standard basis vectors of $\mathbb{R}^{|G|}$ as e_g for $g \in G$ and then write

$$\Delta^G := \Delta^{|G|-1} = \left\{ \sum_{g \in G} x_g e_g \in \mathbb{R}^{|G|} : \forall g \in G, x_g \in [0, 1] \text{ and } \sum_{g \in G} x_g = 1 \right\}$$

which gives us the bijection

$$\mathcal{P}(G) \ni \mu \mapsto \sum_{g \in G} \mu(g) e_g \in \Delta^G.$$

Because of this, I will treat measures $\mu \in \mathcal{P}(G)$ as vectors in $\mathbb{R}^{|G|}$. Similarly, a function $f : G \rightarrow \mathbb{R}$ is equivalently a vector $f \in \mathbb{R}^{|G|}$. This gives us that

$$\int_G f d\mu = \sum_{g \in G} \mu(g) f(g) = \mu \cdot f$$

where $\cdot$ is the standard Euclidean dot product.

Going forward, we will assume that G acts freely on X , which means that $\mu_x = \langle \mu_x \rangle_y$ for any weighted frame $\mu_\bullet$ and $x, y \in X$ since $G_y = \{e\}$. In particular, for some convergent sequence $x_n \rightarrow x \in X$, this means $\langle \mu_{x_n} \rangle_x \rightarrow \langle \mu_x \rangle_x$ if and only if $\mu_{x_n} \rightarrow \mu_x$. But this means that for all $f \in \mathbb{R}^{|G|}$, $\mu_{x_n} \cdot f \rightarrow \mu_x \cdot f$, and thus $\mu_{x_n} \rightarrow \mu_x$ under the usual Euclidean topology. Hence, continuity of the weighted frame $\mu_\bullet$ is equivalent to continuity of the map $x \mapsto \mu_x$. In summary, a continuous weighted frame for G acting on X is simply a continuous map from X into Δ^G when G is finite and acts freely on X .

We also want (weakly) equivariant weighted frames, which in this case are weighted frames such that $\mu_{hx}(g) = \mu_x(h^{-1}g)$ (equivariant and weakly equivariant frames are the same when G acts freely). Now let $\varphi : G \rightarrow \text{Sym}(G)$ be the injective group homomorphism $\varphi(g) = g^{-1} \cdot -$ where $\text{Sym}(G)$ is the symmetric group on G , so we want $\mu_{hx}(g) = \mu_x(\varphi(h)g)$. But $\text{Sym}(G)$ acts naturally on Δ^G via permutation, so an equivariant weighted frame is really just a G -equivariant map from X to Δ^G where G acts on $x \in \Delta^G$ by $g \cdot x := \varphi(g)x$. In this setting, we have the following theorem.

Theorem 2.2.13. *A map $f : X \rightarrow \Delta^G$ is G -equivariant if and only if there is some map $\alpha : X \rightarrow [0, 1]$ such that $\sum_{g \in G} \alpha(gx) = 1$ for all $x \in X$ and*

$$f(x) = \sum_{g \in G} \alpha(g^{-1}x) e_g.$$

Proof. Note that the reverse direction is trivial, so we just prove the forward direction. Since $|G| < \infty$, we must be able to write $f(x) = \sum_{g \in G} f_g(x) e_g$ for continuous $f_g : X \rightarrow [0, 1]$ with $\sum_{g \in G} f_g(x) = 1$ at all $x \in X$. Then the equivariance of f gives us

$$\sum_{g \in G} f_g(hx) e_g = f(hx) = h f(x) = h \sum_{g \in G} f_g(x) e_g = \sum_{g \in G} f_{h^{-1}g}(x) e_g.$$

Hence, for each $g, h \in G$ and $x \in X$ we have $f_g(hx) = f_{h^{-1}g}(x)$, so we can take $\alpha := f_e$. Observe that α is certainly continuous, and we conclude by computing

$$\sum_{g \in G} \alpha(g^{-1}x) = \sum_{g \in G} f_e(g^{-1}x) = \sum_{g \in G} f_g(x) = f(x).$$

□

We would generally like to construct equivariant frames that are computationally efficient, so we make the following definition.

Definition 2.2.14. *The cardinality of a weighted frame $\mu_\bullet$ is*

$$|\mu_\bullet| := \sup_{x \in X} \min \left\{ n \in \mathbb{N} : \exists g_1, \dots, g_n \in G, \alpha_1, \dots, \alpha_n \in \mathbb{R}, \mu_x = \sum_{i \leq n} \alpha_i \delta_{g_i} \text{ and } \sum_{i \leq n} \alpha_i = 1 \right\}.$$

In other words, $|\mu_\bullet|$ is the minimum number of Dirac measures needed to represent μ_x for all $x \in X$. We take $|\mu_\bullet| = \infty$ if μ_x cannot be written as a finite sum of Dirac measures for some $x \in X$.

And we can make a similar definition for canonicalization frames.

Definition 2.2.15. *The cardinality of a weighted canonicalization frame $\nu_\bullet$ is*

$$|\nu_\bullet| := \sup_{x \in X} \min \left\{ n \in \mathbb{N} : \exists x_1, \dots, x_n \in X, \alpha_1, \dots, \alpha_n \in \mathbb{R}, \mu_x = \sum_{i \leq n} \alpha_i \delta_{x_i} \text{ and } \sum_{i \leq n} \alpha_i = 1 \right\}.$$

We take $|\nu_\bullet| = \infty$ if ν_x cannot be written as a finite sum of Dirac measures for some $x \in X$.

Note that the frames constructed in theorem 2.1.33 satisfy $|\mu_\bullet| = |\nu_\bullet|$ when G acts freely on X , and this continues to hold when G is infinite.

Observe then that $|\mu_\bullet| = k$ means that the image of $x \mapsto \mu_x$ lives in the $(k-1)$ -skeleton of Δ^G , and notably is not fully contained in the $(k-2)$ -skeleton. Hence, we have the following proposition.

Proposition 2.2.16. *Let X be a free G -space for a finite group G . Then there exists a continuous equivariant weighted frame $\mu_\bullet : X \rightarrow \mathcal{P}(G)$ with $|\mu_\bullet| = k$ if and only if there exists a G -equivariant map $f : X \rightarrow_G (\Delta^G)^{(k-1)}$.*

We now make an observation about how G acts on $(\Delta^G)^{(k)}$ for various k .

Proposition 2.2.17. *G acts freely on $(\Delta^G)^{(k)}$ if and only if G has no nontrivial subgroup H such that $|H| < k$.*

Proof. First, assume there is some subgroup $H \subseteq G$ with $1 < |H| < k$ and fix some $h \in H$. Observe that $\sum_{g \in H} \frac{1}{|H|} e_g \in (\Delta^G)^{(k)}$ but

$$h \cdot \sum_{g \in H} \frac{1}{|H|} e_g = \sum_{g \in H} \frac{1}{|H|} e_{h^{-1}g} = \sum_{g \in H} \frac{1}{|H|} e_g$$

so $\sum_{g \in H} \frac{1}{|H|} e_g$ has nontrivial stabilizer (namely, its stabilizer is H).

Conversely, assume that there is some $x = \sum_{g \in G} x_g e_g \in (\Delta^G)^{(k)}$ with nontrivial stabilizer. Note that G acts on x via cyclic permutations of the nonzero coordinates, of which there are less than k , so we must have that $|G_x| < k$. Hence, G_x is exactly the nontrivial subgroup we desire. $\square$

Theorem 2.2.18 (Dold's Theorem [Dol83], as in Theorem 6.2.6 of [Mat07]). *Let G be a nontrivial finite group, let X be an n -connected G -space, and let Y be a free G -space of dimension at most n . Then there are no G -equivariant maps from X to Y .*

Theorem 2.2.19. *Let G be a nontrivial finite group, let p be the least prime divisor of $|G|$, and let X be an n -connected G -space for $n \geq p$. Then there is no robust frame $\mu_\bullet : X \rightarrow \mathcal{P}(G)$ with $|\mu_\bullet| < p$.*

Proof. Since p is the least prime divisor of $|G|$, there can be no subgroups H of G with $|H| < p$ and hence G acts freely on $(\Delta^G)^{(k)}$ for any $k \leq p$ by proposition 2.2.17. Clearly $\dim(\Delta^G)^{(k)} = k \leq p \leq n$, so by theorem 2.2.18 we then have that there can be no G -equivariant map from X to $(\Delta^G)^{(k)}$. Finally, by proposition 2.2.16 we conclude that there can be no robust frame $\mu_\bullet : X \rightarrow \mathcal{P}(G)$ with $|\mu_\bullet| = k + 1$ for any $1 \leq k \leq p$. $\square$

Corollary 2.2.20. *If X is simply connected then there are no nontrivial robust frames when $G = \mathbb{Z}_p$ for p prime.*

We now return to an idea from chapter 1 that finds new use.

Proposition 2.2.21. *Let X be a G -space such that X/G is locally compact and Hausdorff. Furthermore, let $\{U_i\}_{i \leq n}$ for $n \in \mathbb{N}$ be a collection of open sets in X/G such that $X/G = \cup_{i \leq n} U_i$ with local sections $s_i : U_i \rightarrow X$ of the quotient $q : X \rightarrow X/G$. Then there exists a robust canonicalization frame $\nu_\bullet$ such that $|\nu_\bullet| \leq n$.*

Proof. Since X/G is locally compact and Hausdorff, there exists a partition of unity $\{\varphi_i\}_{i \leq n}$ subordinate to $\{U_i\}_{i \leq n}$. We then define $\nu_\bullet : X \rightarrow \mathcal{P}(X)$ by

$$\nu_x = \sum_{i \leq n} \varphi_i(q(x)) \delta_{s_i(q(x))},$$

noting that $\varphi_i(q(x)) = 0$ when the expression $s_i(q(x))$ doesn't make sense so we can ignore this case. Observe that this is well-defined since $\sum_{i \leq n} \varphi_i(q(x)) = 1$ because $\{\varphi_i\}_{i \leq n}$ is a partition of unity. Additionally, $\nu_\bullet$ is a canonicalization frame by construction since it only depends on $q(x)$, and for any continuous map $f : X \rightarrow \mathbb{R}$ we have

$$\int_X f d\nu_x = \sum_{i \leq n} \varphi_i(q(x)) f(s_i(q(x)))$$

which is continuous as a composition of continuous functions, so $\nu_\bullet$ is a robust frame. Lastly, by construction $|\nu_\bullet| \leq n$ so we are done. $\square$

2.2.3 Canonicalization Frame Constructions

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